

Cell-Cell communication in spatial omics

Francesca Drummer

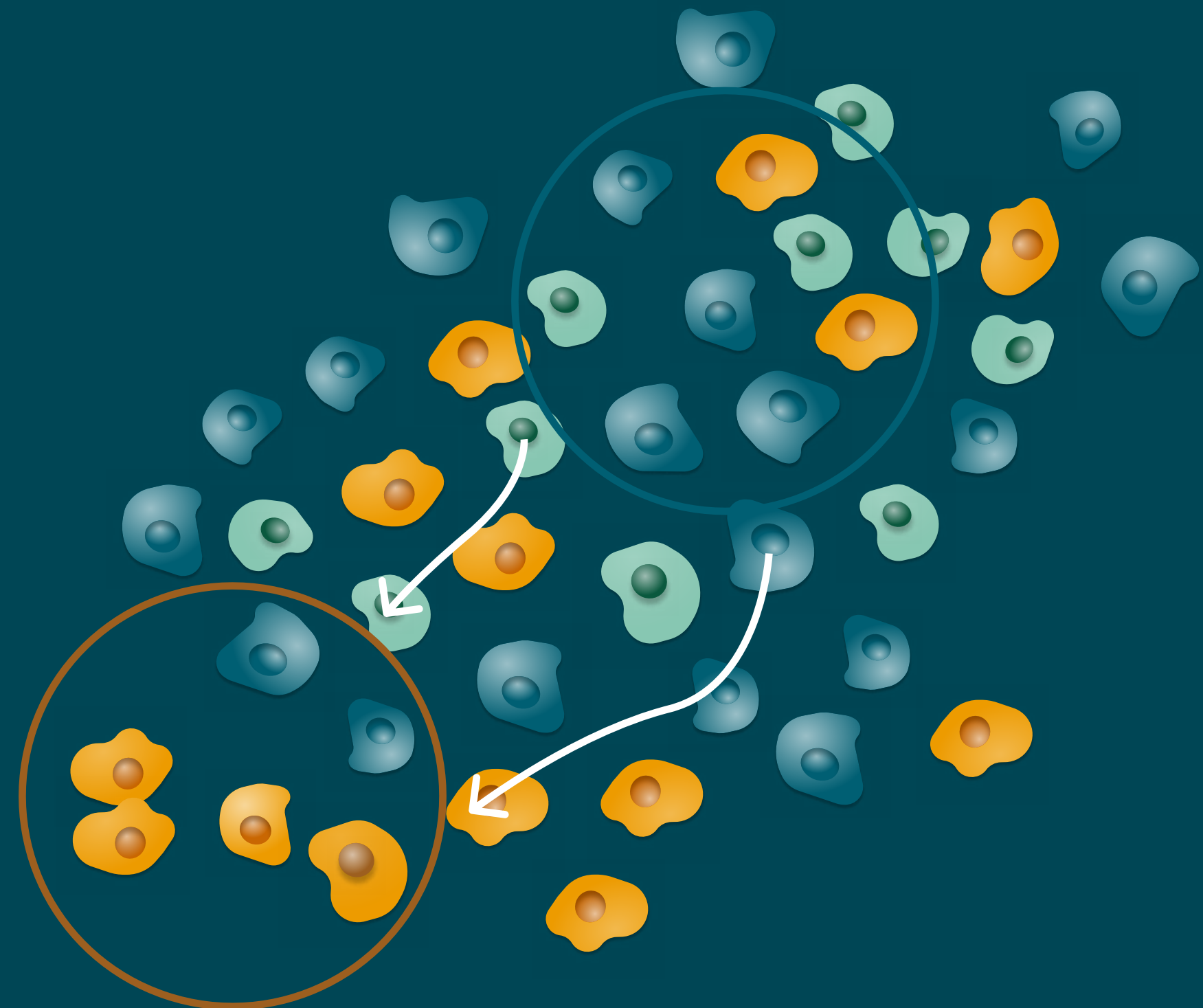
Theis lab - Institute of Computational Biology, Helmholtz Munich

09.10.2026 - ELIXIR Spatial Omics workshop

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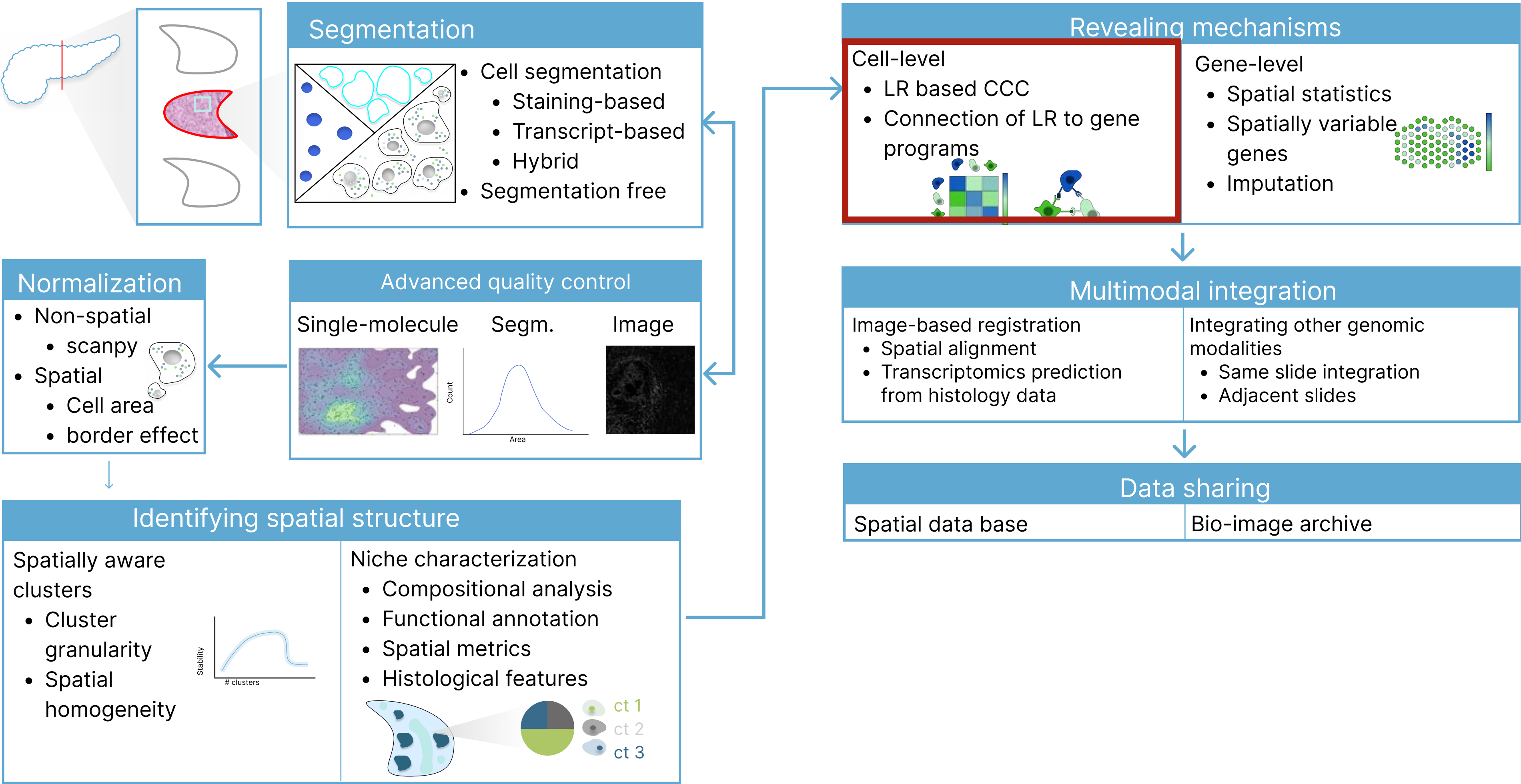
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[GitHub](#) FrancescaDr



slides inspired by Y. Saez, ELIXIR SCO course 2025 and scverse GSCN course 2026

Hitchhikers guide for imaging-spatial transcriptomics



Sara Jimenez

Now: @NVIDIA
Prev.: Postdoc @Theislab



Fabian J. Theis

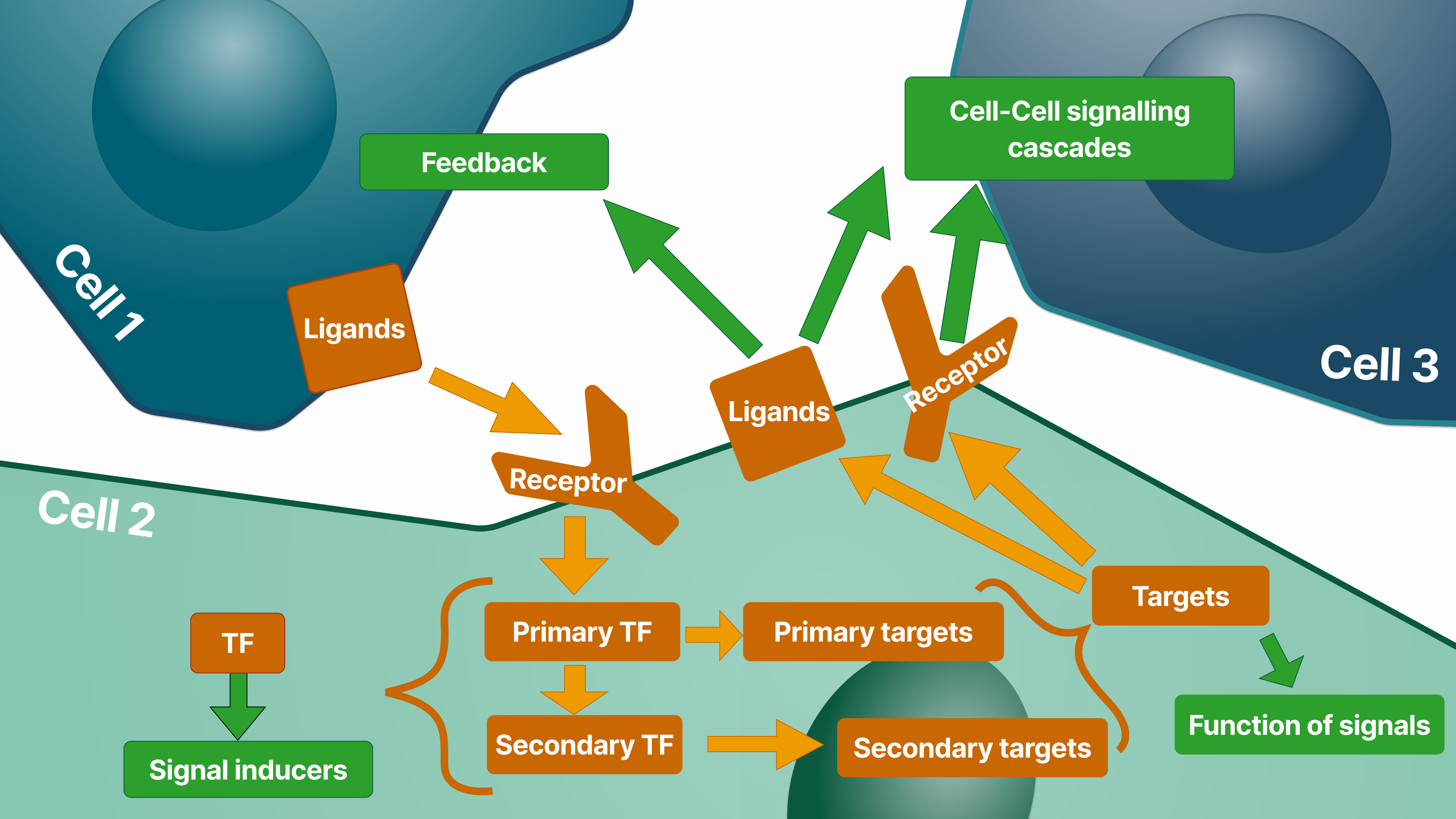
PI @ Helmholtz Munich

Naveed Ishaque
+ many other contributors



What types of data would we need to study functional cell-cell communication?

Functional communication meaning a signalling event has a effect



Ligands & receptors:
Proteomics, metabolomics,
Imaging, CITE-seq,...

Dynamics:
time series, perturbations, (e.g.
perturb-seq, CRISPr, ligand
treatments), intravital imaging,...

**Cell signalling
cascades**

Feedback

intER-cellular signalling

Ligands

Ligands

Receptor

Receptor

Spatial location:
Spatial proteomics,
spatial transcriptomics,
spatial metabolomics,...

Gene regulation:
scRNA-seq, scATAC-
seq, CHIP-seq,...

Function of signals

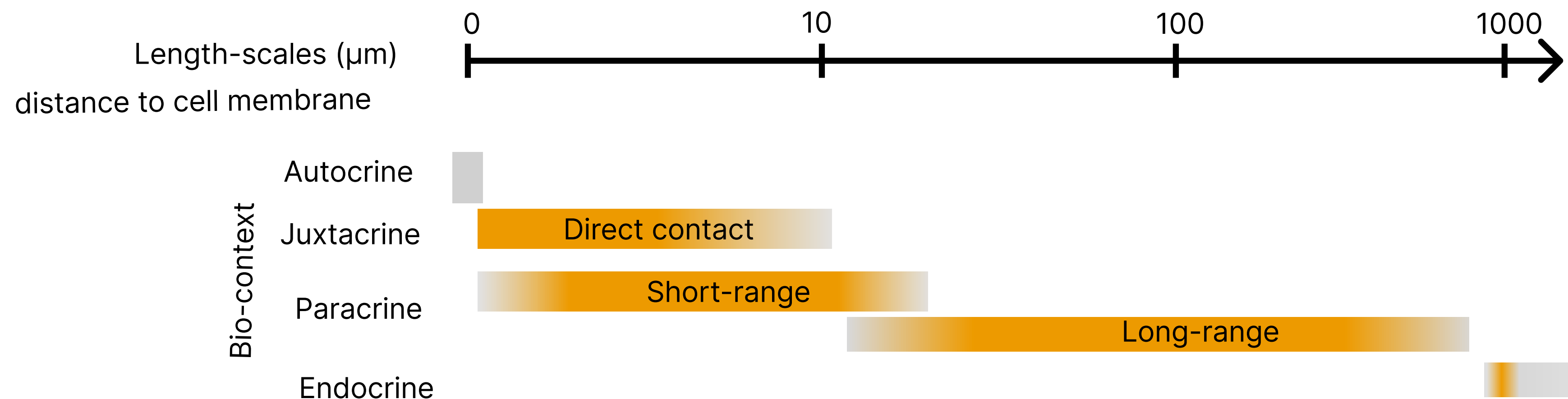
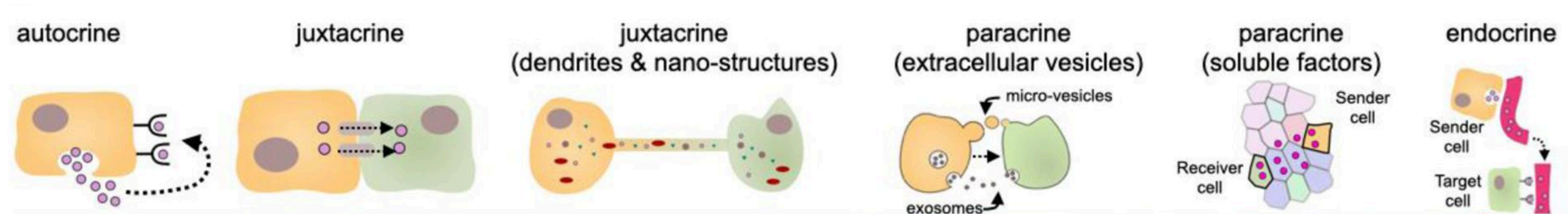
TF
Signal inducers

Primary **intRA-cellular signalling** **ts**

Secondary TF

Secondary targets

Cells interact across different length scales



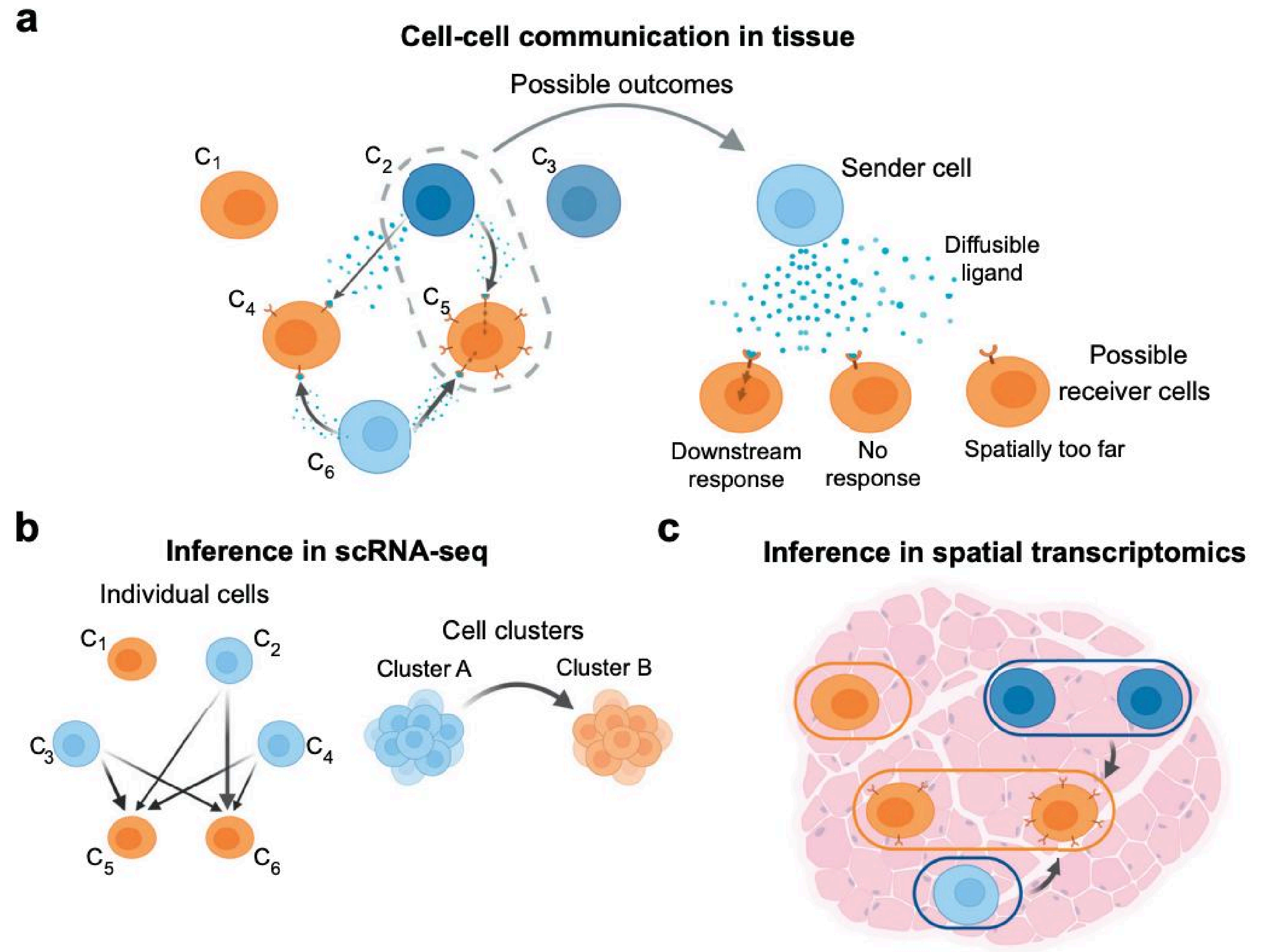
Ligand-Receptor signalling

i.e. morphogen, cytokine signalling

Most current methods focus on CCC inference from transcriptomics data

**Who says what to whom, how, why,
when and where ?**

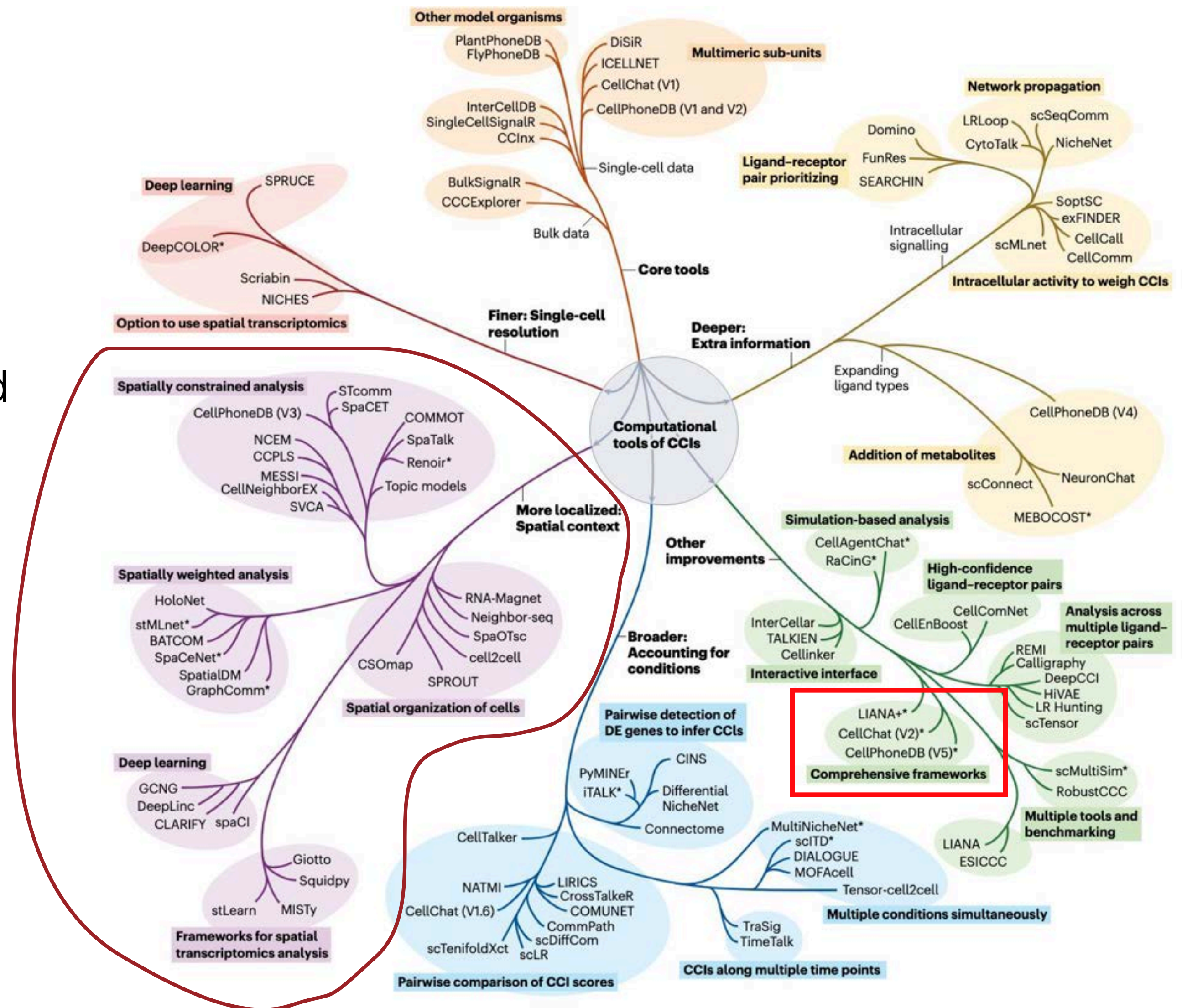
Principles of cell-cell inference



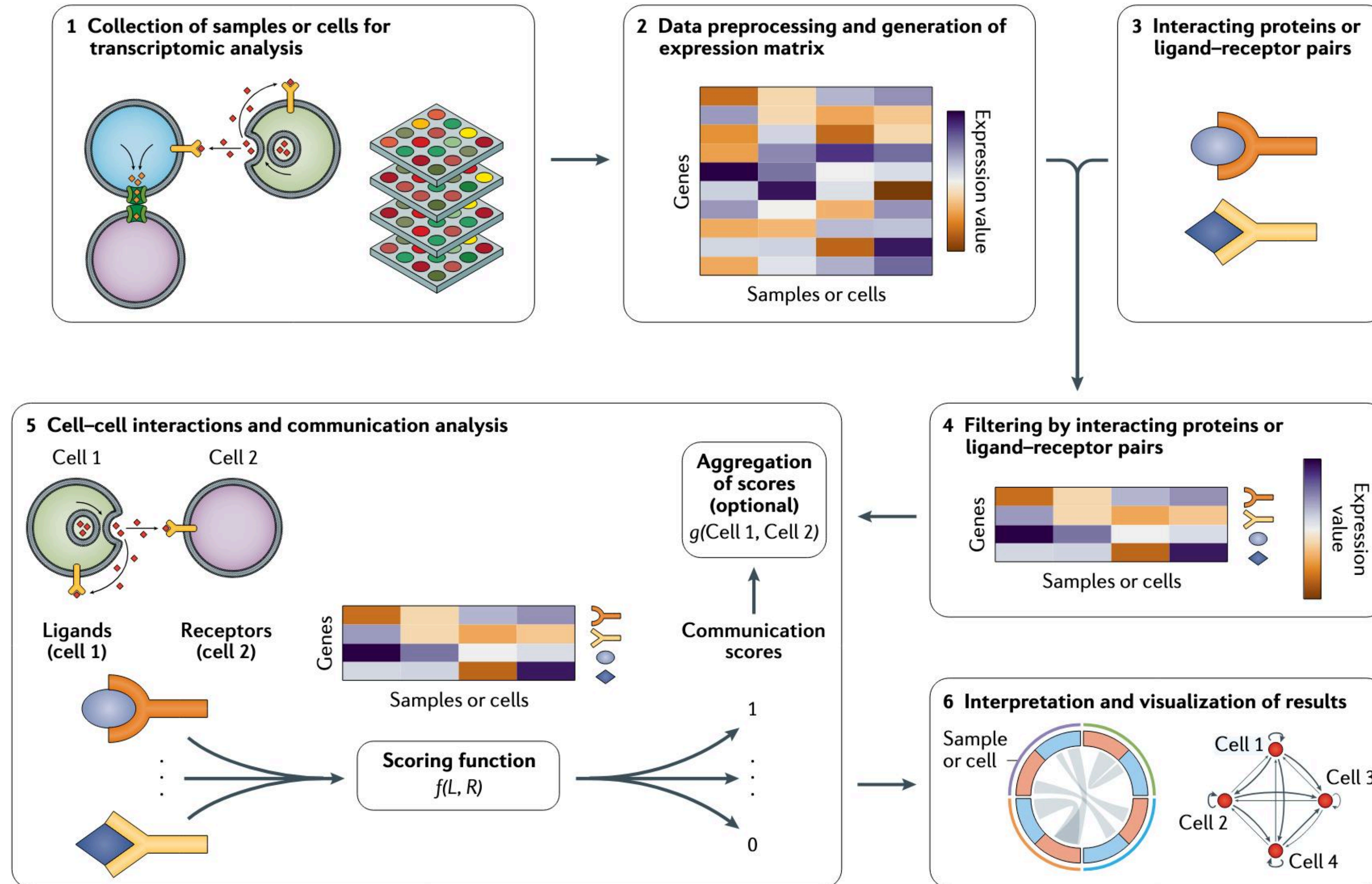
Computational tools for inferring CCI have evolved from a 'root' of core tools

From this root, methods have become more specialized to address specific opportunities.

Most tools are developed for disassociated CCC inference (i.e. CellChat, CellPhoneDB, **LIANA+**)

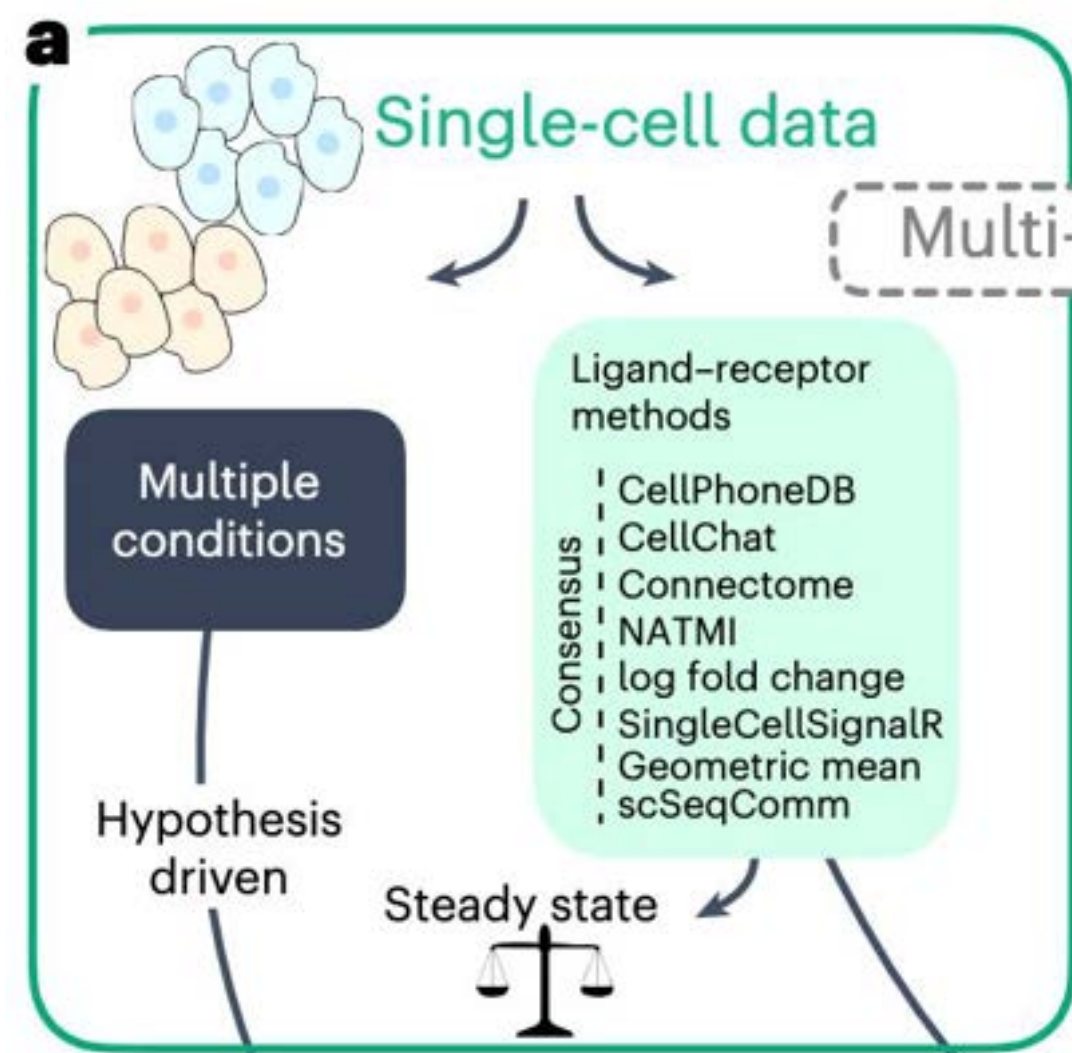


Key steps for Ligand-Receptor methods





LIANA+ is a scalable framework that integrates methods to infer CCC

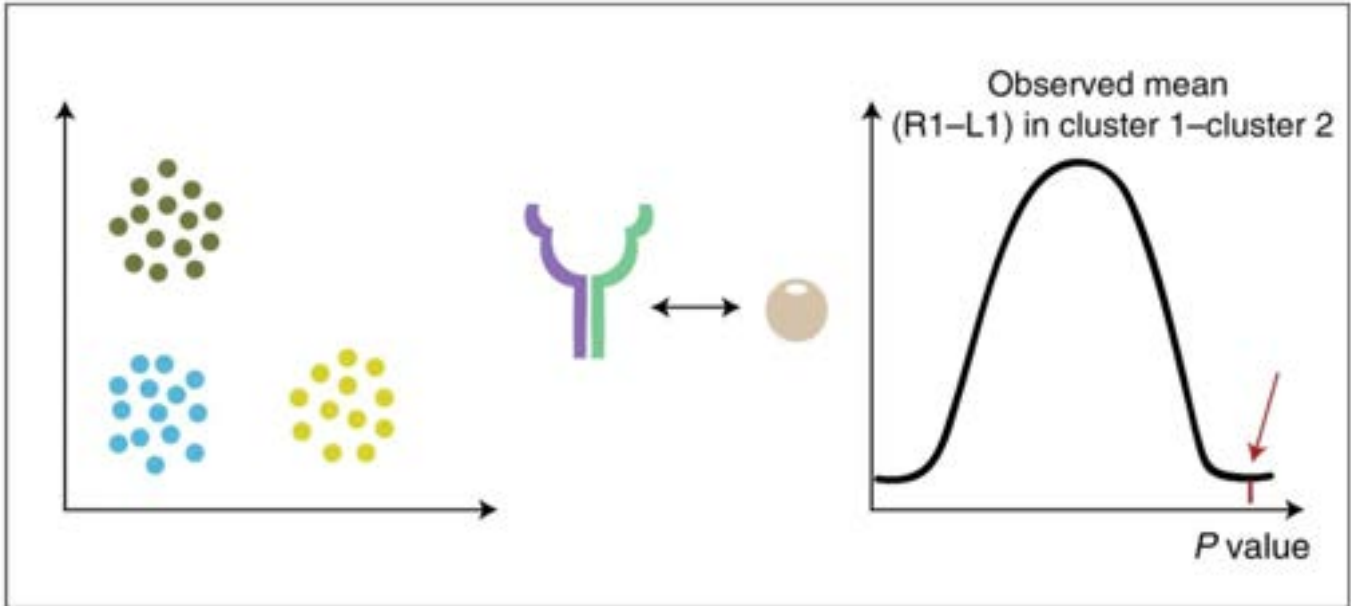
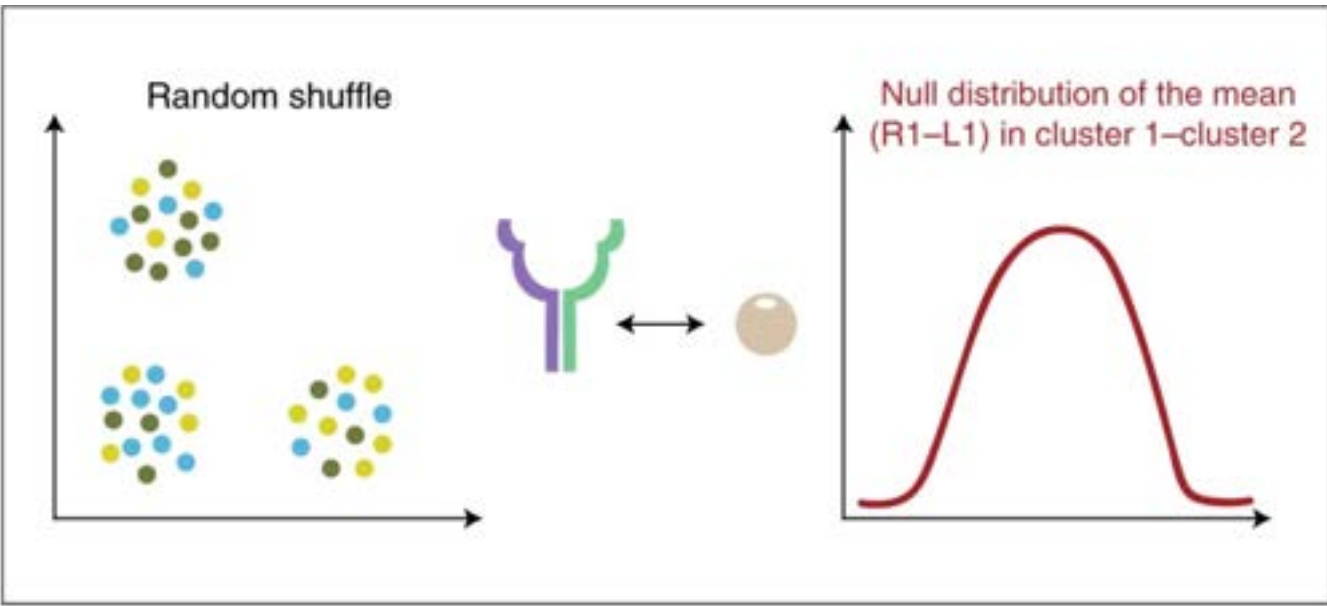


expression
permutation-based
tools

```
li.mt.show_methods()
```

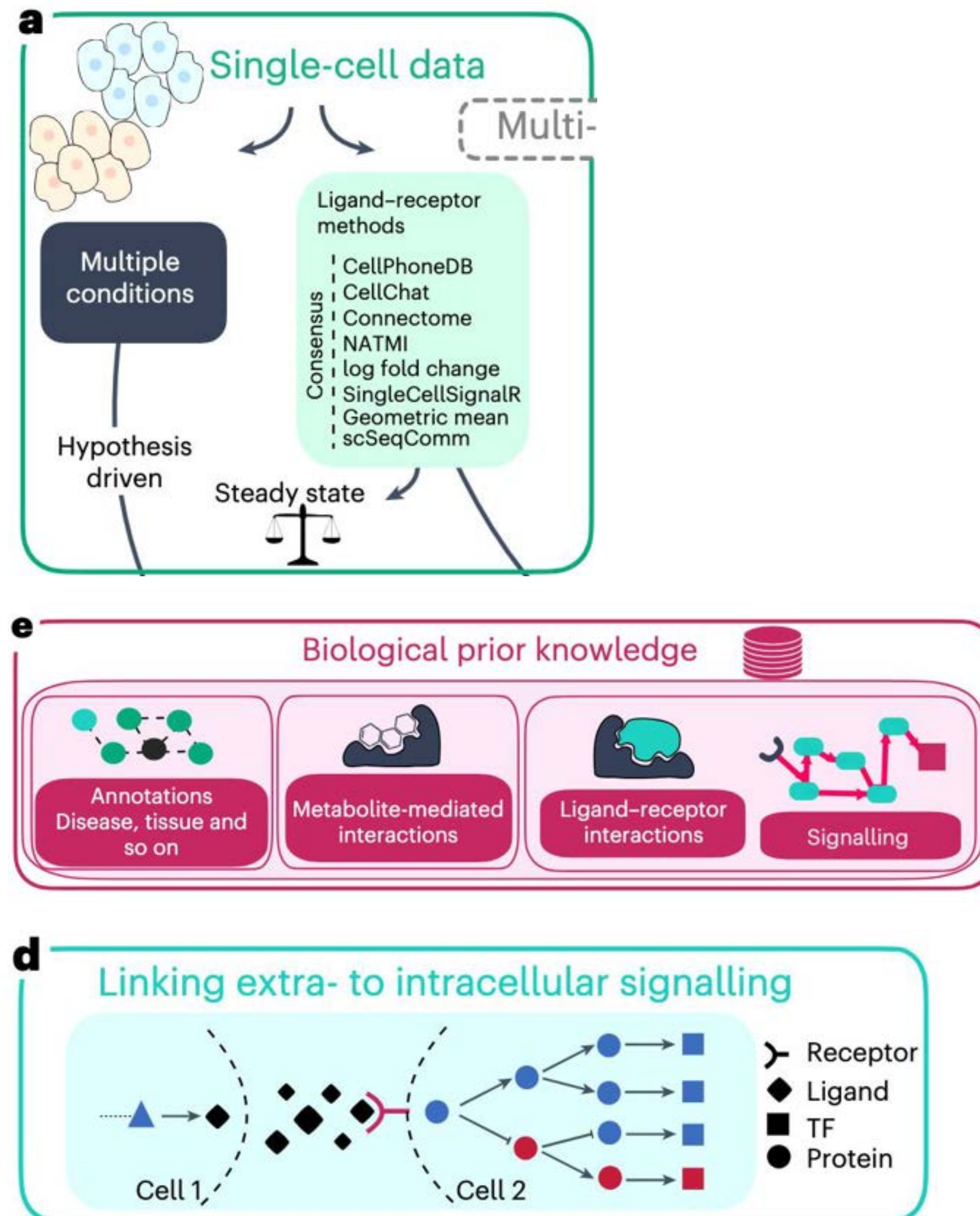
	Method Name	Magnitude Score	Specificity Score	Reference
0	CellPhoneDB	lr_means	cellphone_pvals	Efremova, M., Vento-Tormo, M., Teichmann, S.A....
0	Connectome	expr_prod	scaled_weight	Raredon, M.S.B., Yang, J., Garritano, J., Wang...
0	log2FC	None	lr_logfc	Dimitrov, D., Türe, D., Garrido-Rodriguez, M....
0	NATMI	expr_prod	spec_weight	Hou, R., Denisenko, E., Ong, H.T., Ramilowski,...
0	SingleCellSignalR	lrscore	None	Cabello-Aguilar, S., Alame, M., Kon-Sun-Tack, ...
0	Rank_Aggregate	magnitude_rank	specificity_rank	Dimitrov, D., Türe, D., Garrido-Rodriguez, M....
0	Geometric Mean	lr_gmeans	gmean_pvals	CellPhoneDBv2's permutation approach applied t...
0	scSeqComm	inter_score	None	Baruzzo, G., Cesaro, G., Di Camillo, B. 2022. ...
0	CellChat	lr_probs	cellchat_pvals	Jin, S., Guerrero-Juarez, C.F., Zhang, L., Cha...

Example: CellPhoneDB (Interaction mean = (mean ligand in A + mean receptor in B) / 2)



- Be mindful of:
- Assumptions of each method
 - Meaning of the LR score of each method

LIANA+ is a scalable framework that integrates methods to infer CCC



For example, **NicheNet** (Browaeys, R., Saelens, W. & Saeys, Y. NicheNet: modeling intercellular communication by linking ligands to target genes. Nat. Methods 17, 159–162 (2020))

Strengths and limitations of LRI based CCC models

- **Strengths**

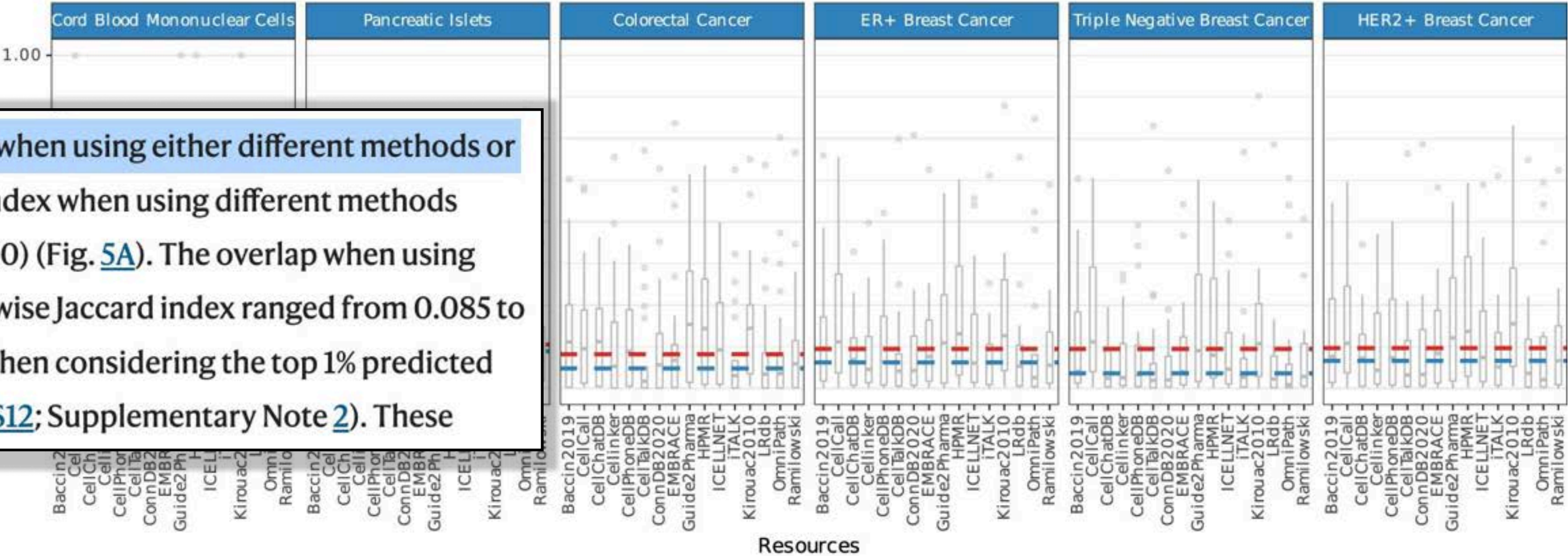
- Different levels of complexity of modelling LR interaction potential based on
- curated databases (e.g. multi-subunit modelling)
- Simple and easy to calculate
- Databases can be expanded, curated,..., to improve models
- Potential for many downstream analyses

- **Limitations**

- Takes LR gene expression as a proxy for functional proteins
- Depends on the curated databases of LR interactions
 - Many tools use different databases → different results
 - Databases are general and not (yet) cell type specific or tissue specific
- Scoring functions and assumptions are different
- Do not take into account downstream functional effects
- No information about spatial proximity

Same Resources with different Methods

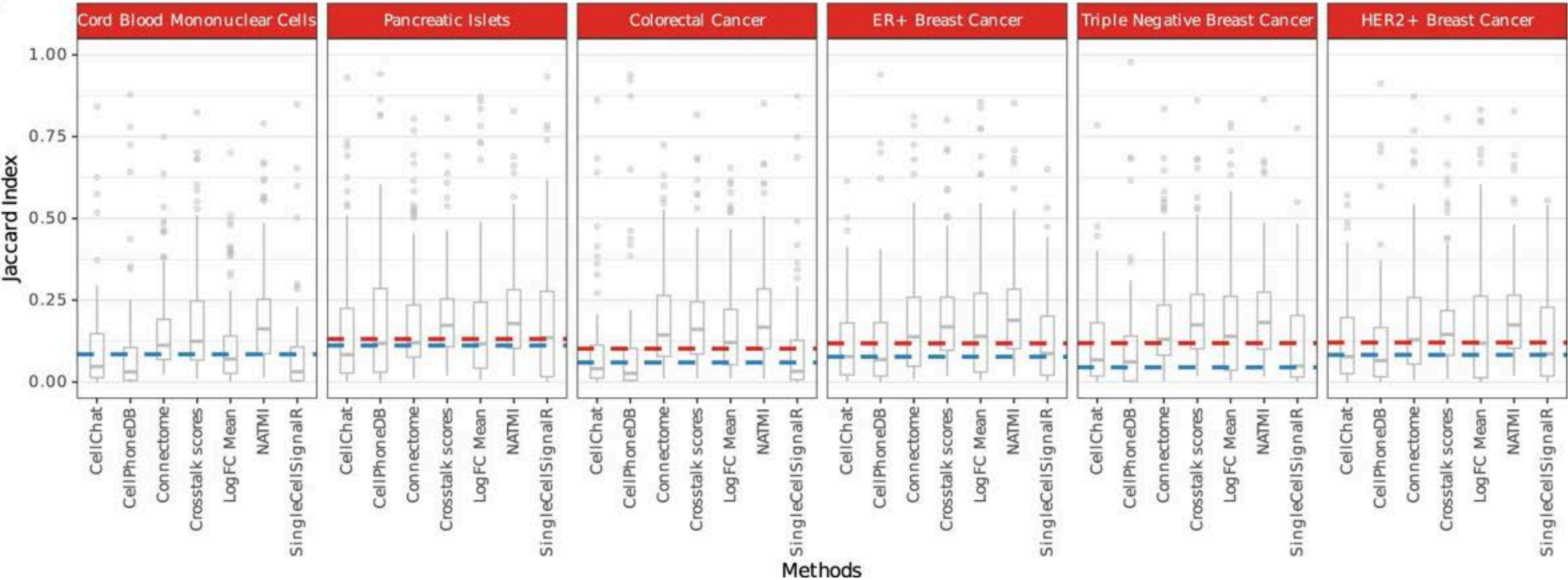
A)



consistently low overlap in the top predicted interactions when using either different methods or different resources (Fig. 5). The median pairwise Jaccard index when using different methods ranged from 0.045 to 0.112 across datasets (median = 0.080) (Fig. 5A). The overlap when using different resources was slightly higher, as the median pairwise Jaccard index ranged from 0.085 to 0.132 (median = 0.119) (Fig. 5B). We found similar results when considering the top 1% predicted interactions instead of the top 1000 (Supplementary Fig. S12; Supplementary Note 2). These

Same Methods with different Resources

B)



Can we reduce inconsistencies by constraining CCC inference with spatial information?

How to integrate spatial information?

How to make any CCC tool spatial?

pre-filtering

pick the cells, then run CCC on them

post-filtering

Run CCC first, then down-weight or filter the ranked LR list by spatial proximity

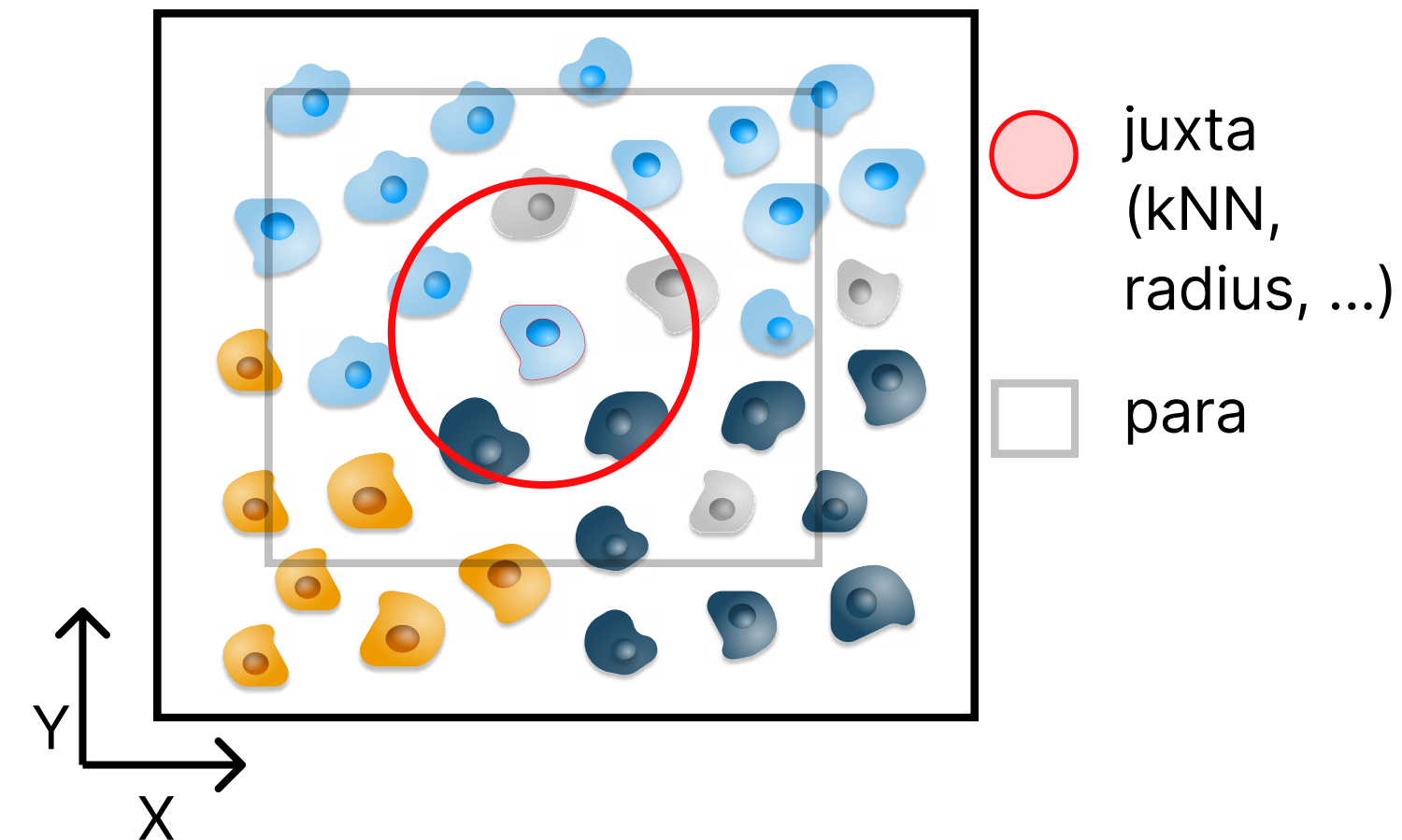
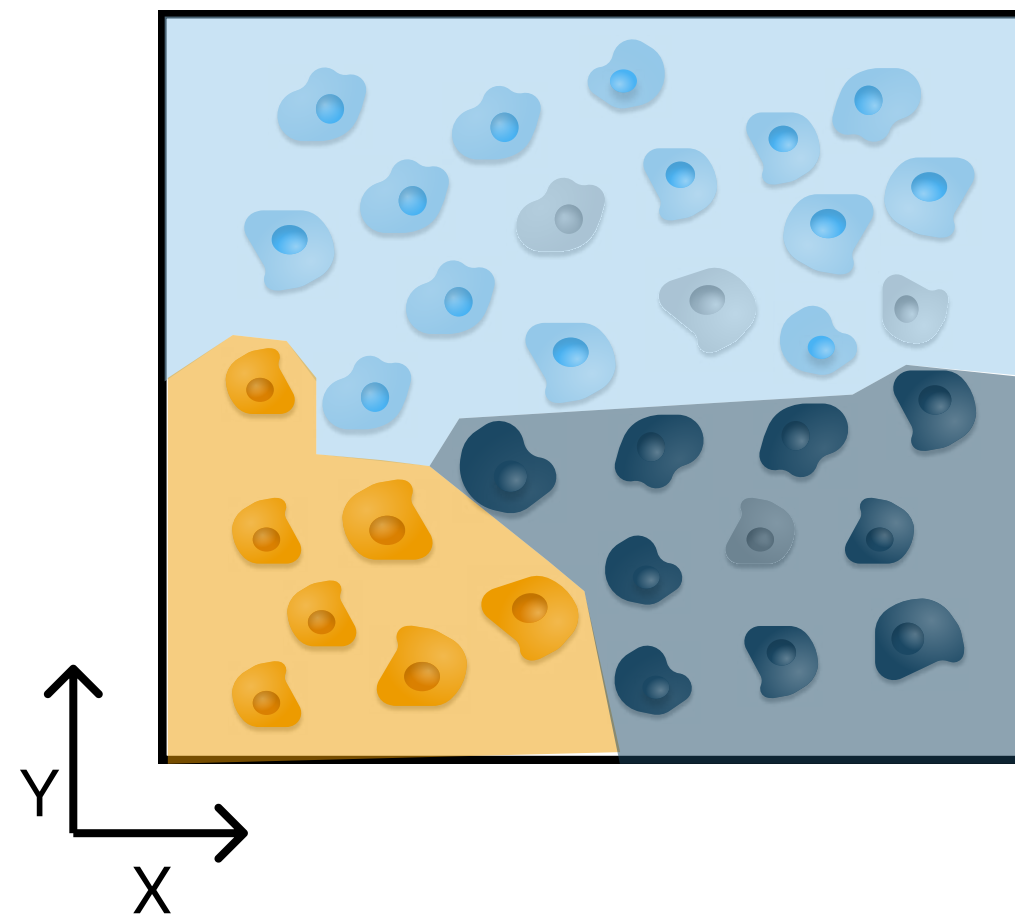
Types of spatially informed CCC analysis

i) fixed distance

(hard constraint)

**regions or microenvironments
(biological prior)**

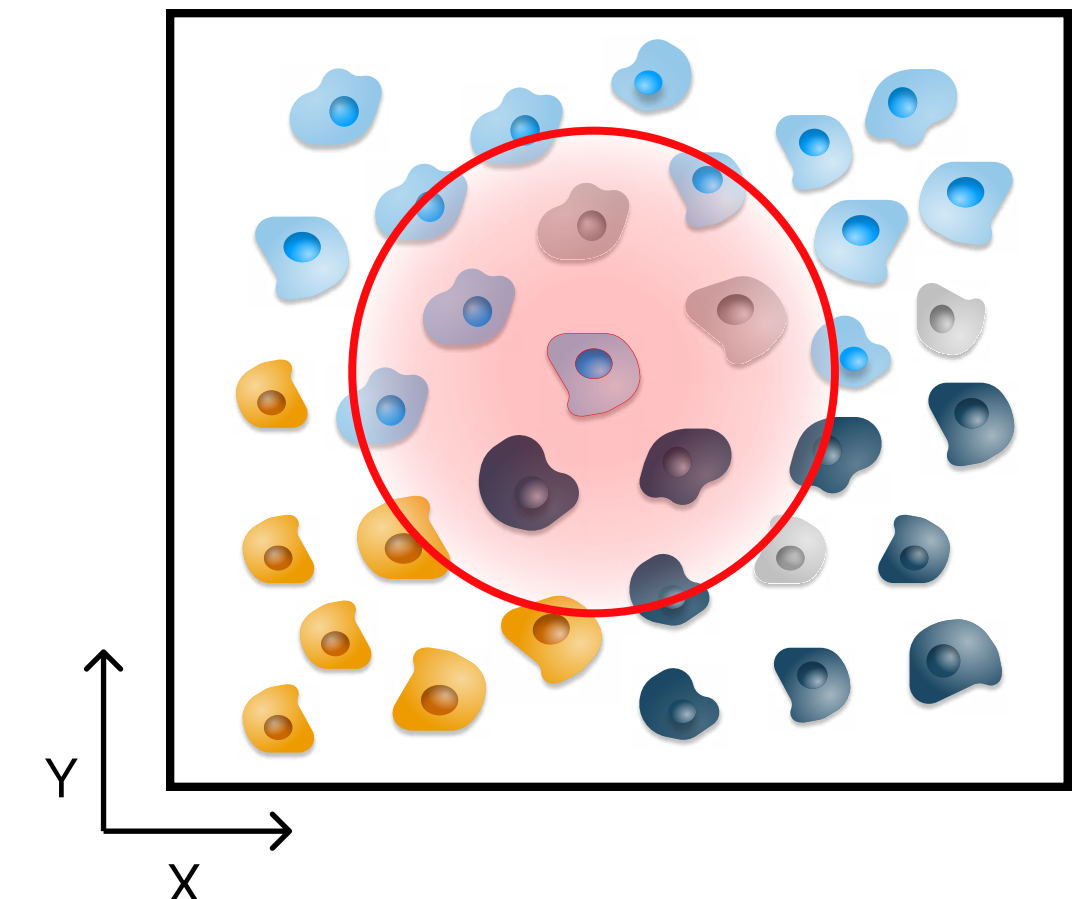
view-specific



ii) interaction decay

(soft constraint)

so nearby cells count more, far cells fade (e.g. hill function, radial kernel)



Roadmap

1

co-localization

Who is near whom, and at what distance?

2

constrained LR inference

Which ligand-receptor pairs,
given proximity?

3

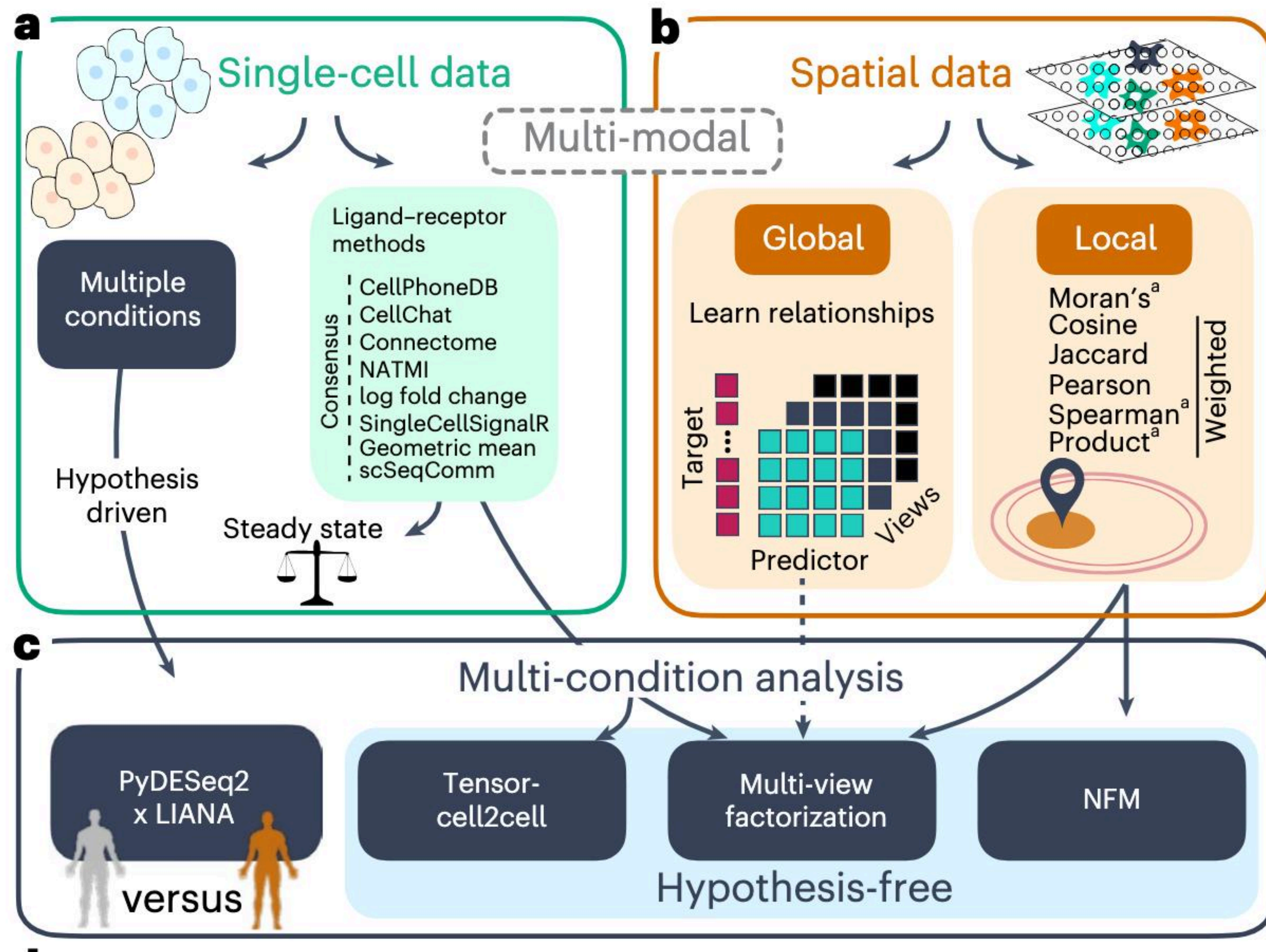
local interaction maps

Where does the interaction
happen locally?

4

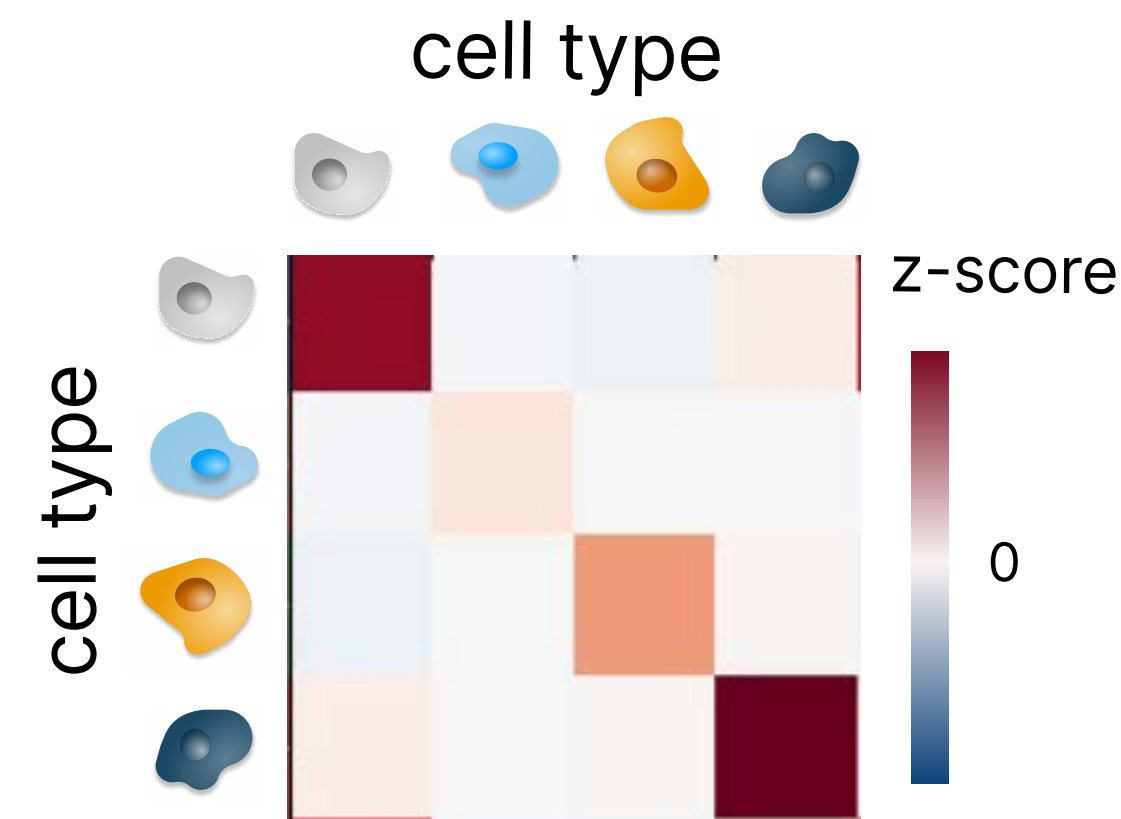
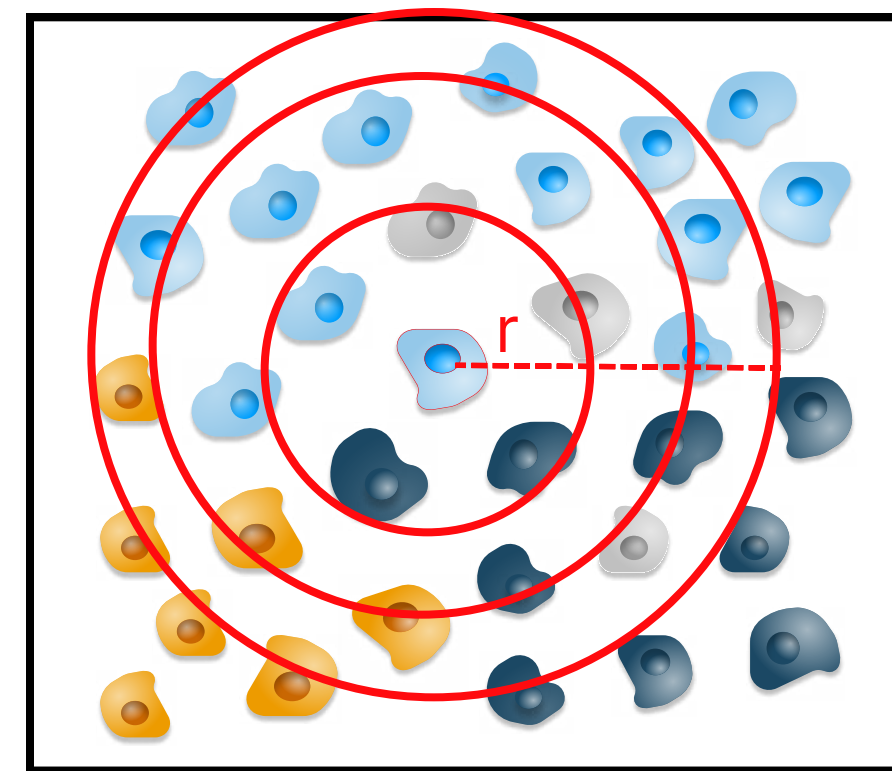
multi-scale models

Across which scales (juxtacrine,
paracrine,...)?



Who is near whom, and at what distance?

Co-localization of cell-types



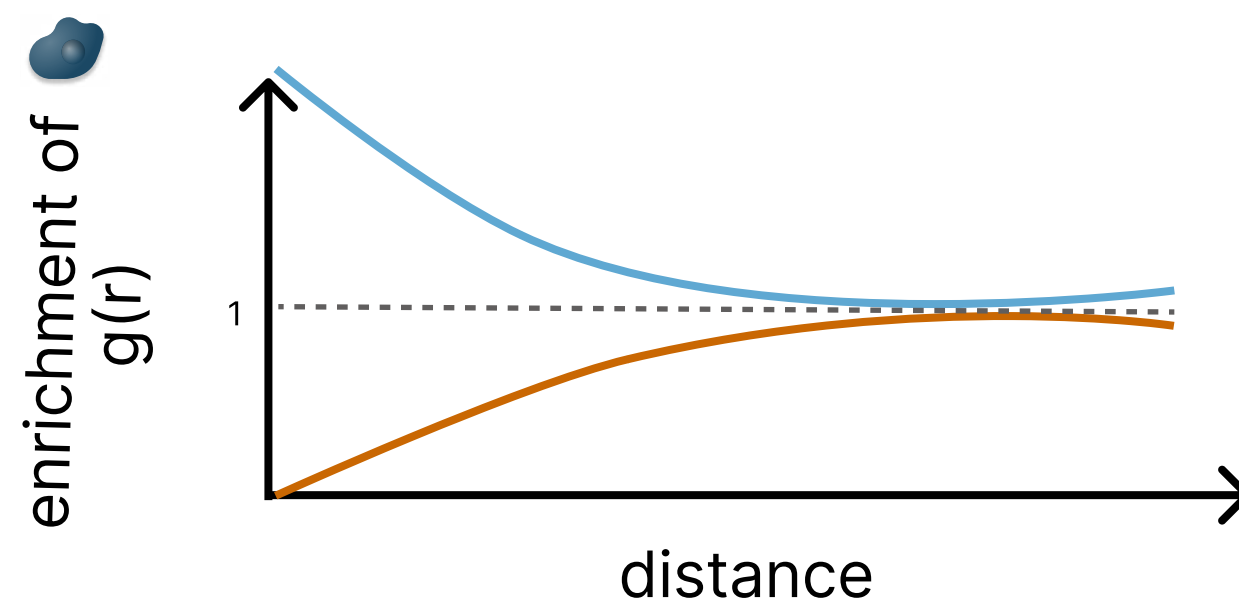
Neighborhood enrichment analysis

1. Construct spatial neighborhood (kNN/radius graph)
2. Permutation z-score for cell type adjacency

e.g.

- squidpy (Palla, G. *et al. Nat Methods* **19**, 171–178 (2022).)
- COZI (Schiller, C. *et al. D. Nat Commun* **17**, 3514 (2026).)

cross-pair correlation function

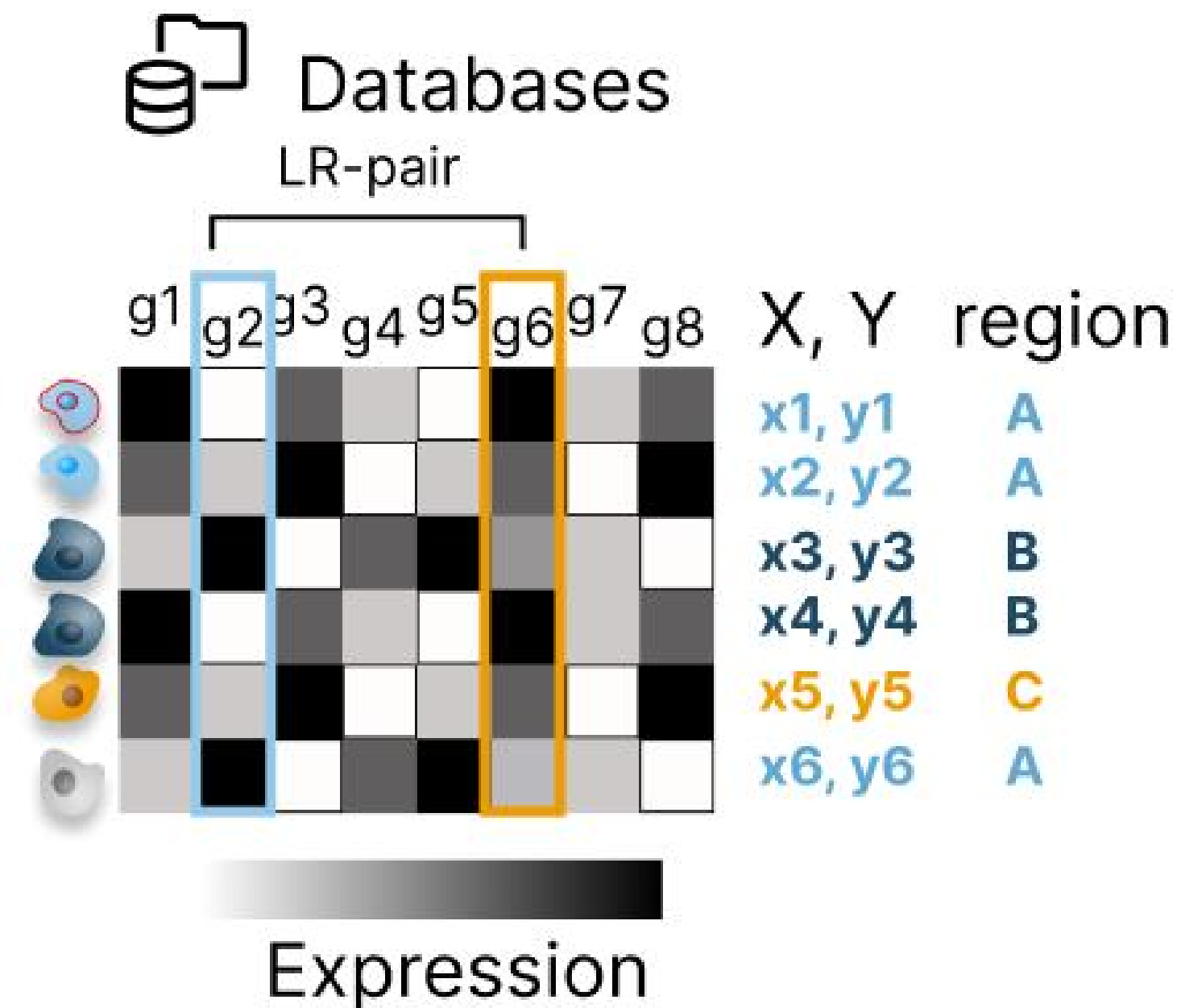


Limitations:

- Hard distance constraint does not consider cell type specific interaction
- No information about mechanistic gene insight

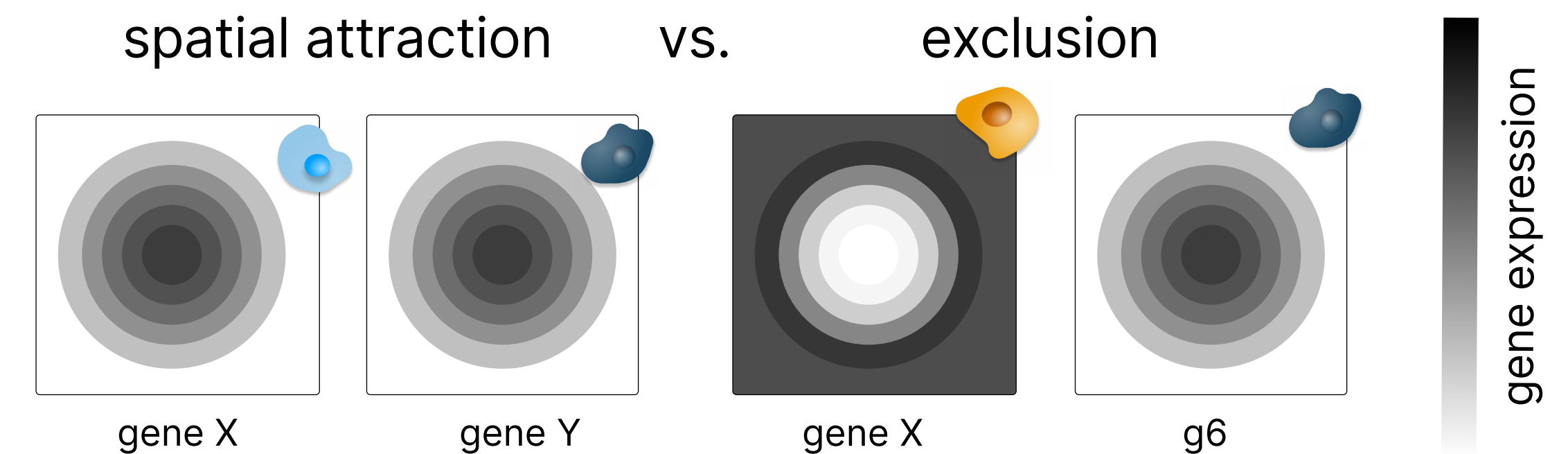
Which ligand-receptor pairs, given proximity?

constrained LR inference



Gene-level

- univariate, bi-variate & multi-variate autocorrelation (e.g. Moran's I, LISA)

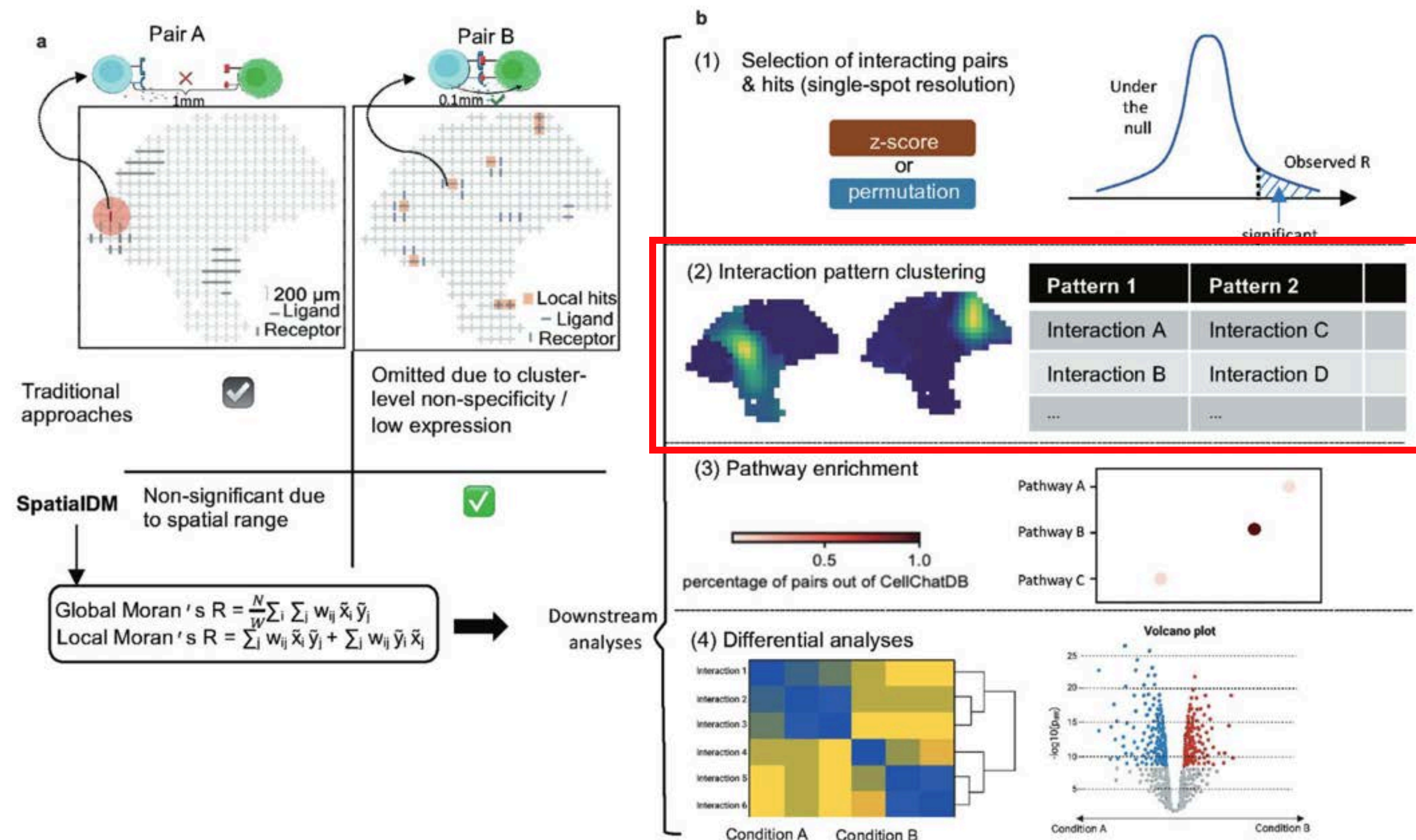


- only test sender-receiver pairs that co-localize (pre-filter; hard constraint)
- proximity weighting (e.g. CellChat v2 adds a distance factor)

Where does the interaction happen locally?

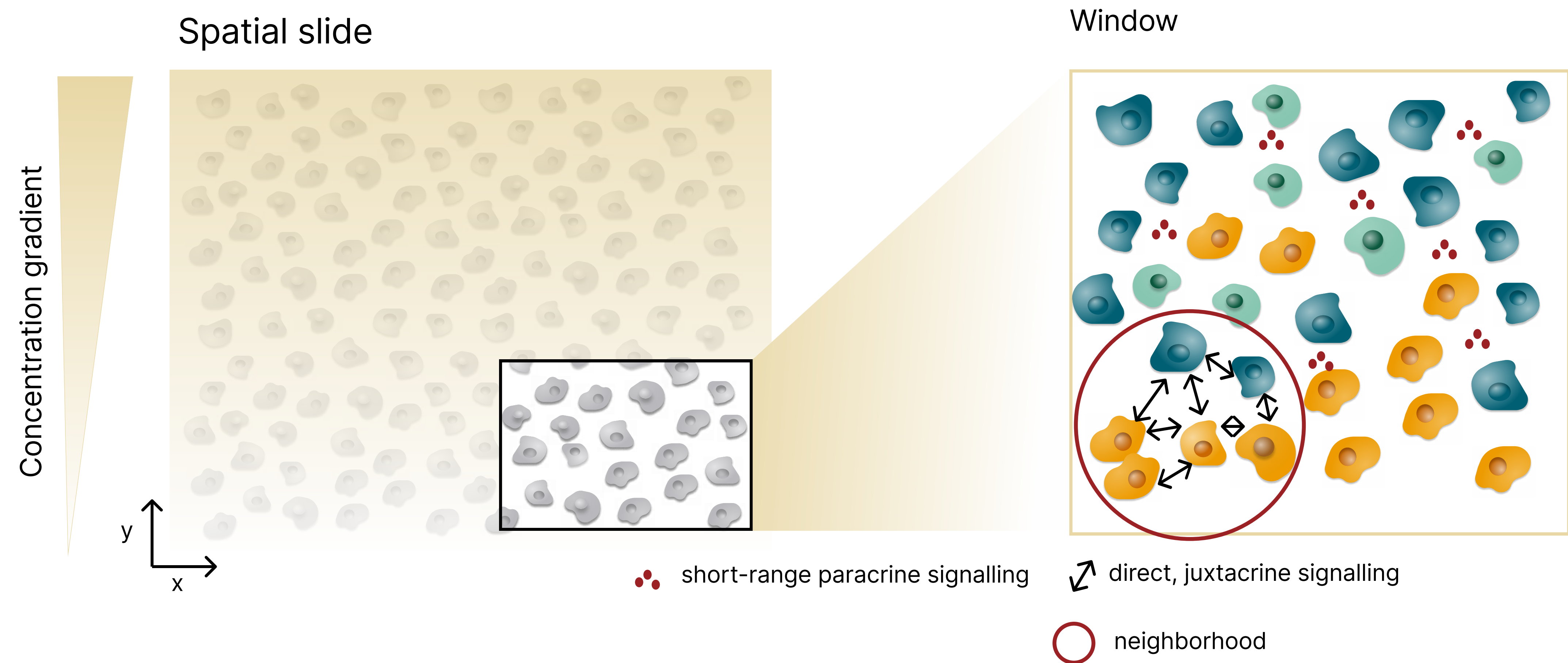
local interaction maps

SpatialDM



In the notebooks we will use LIANA+ Inflow method

Most tools model cell interactions in neighborhoods



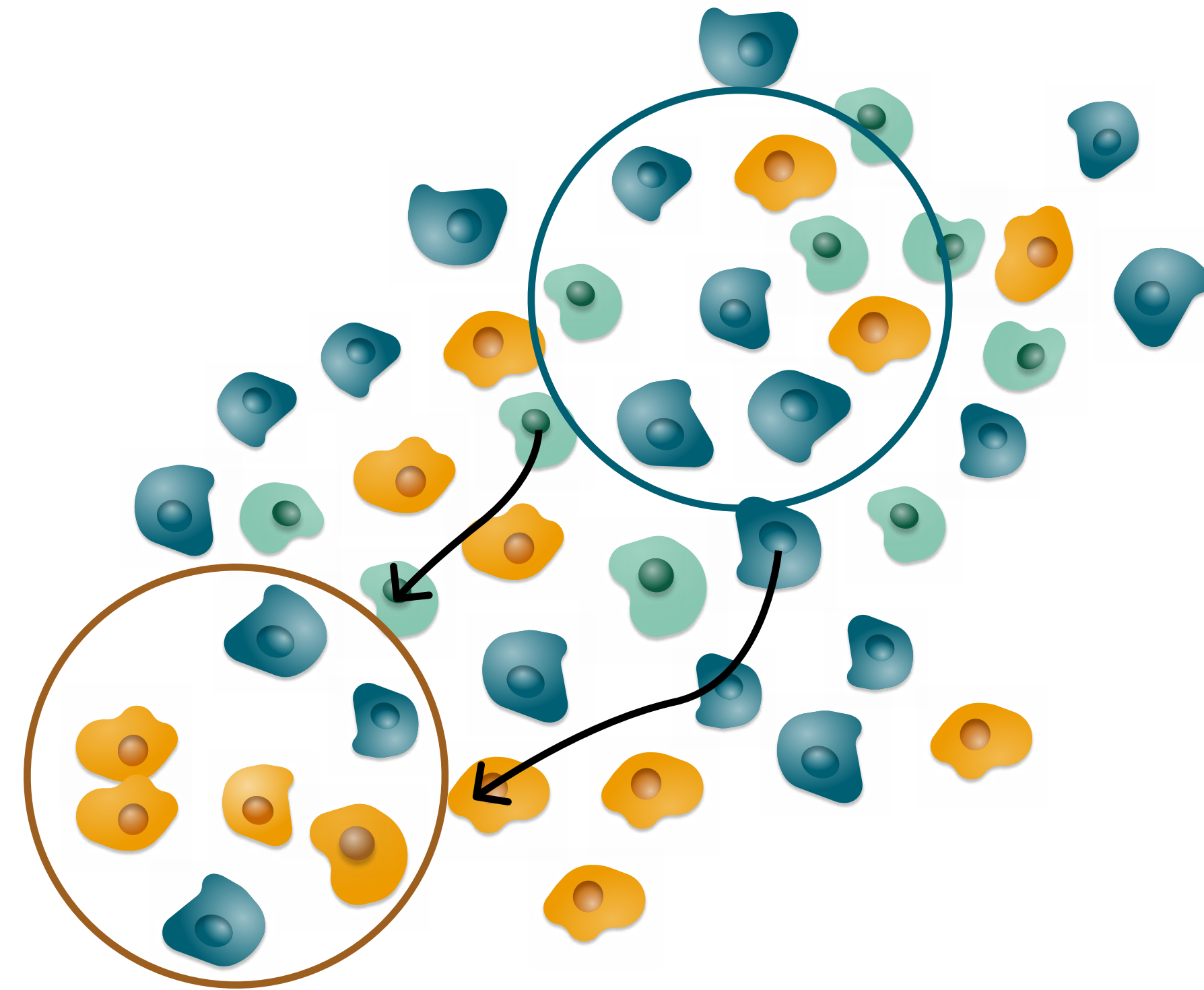
NCEM: Fischer, D. S., Schaar, A. C. & Theis, F. J. Modeling intercellular communication in tissues using spatial graphs of cells. Nature Biotechnology 41, 332–336 (2023).
NicheCompass: Birk, S. et al. Quantitative characterization of cell niches in spatially resolved omics data. Nat Genet 57, 897–909 (2025).

...

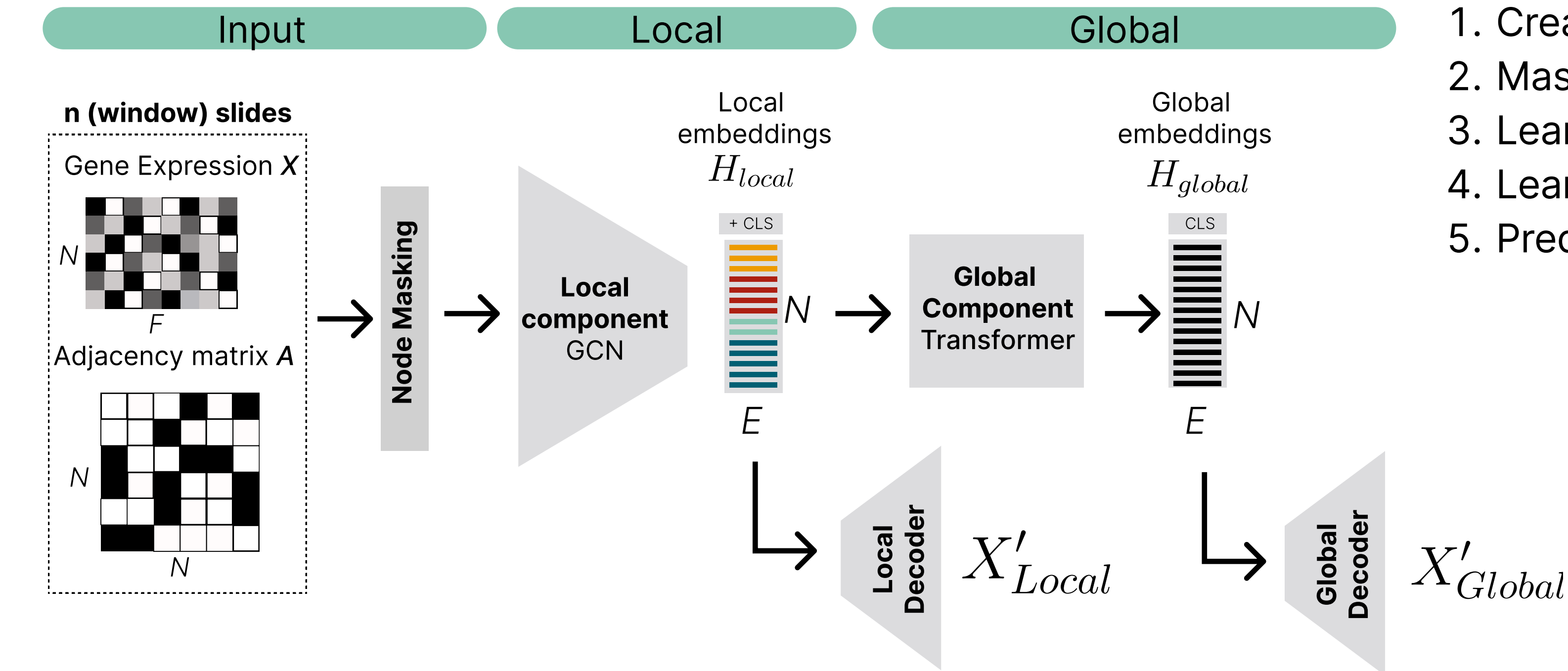
Multi-scale interaction analysis

Drummer, F., Jimenez, S. *et al.* **InterScale reveals multi-scale cellular interaction programs in spatial transcriptomics.**
2026.05.07.723456. <https://doi.org/10.64898/2026.05.07.723456> (2026) - under revisions at MSB

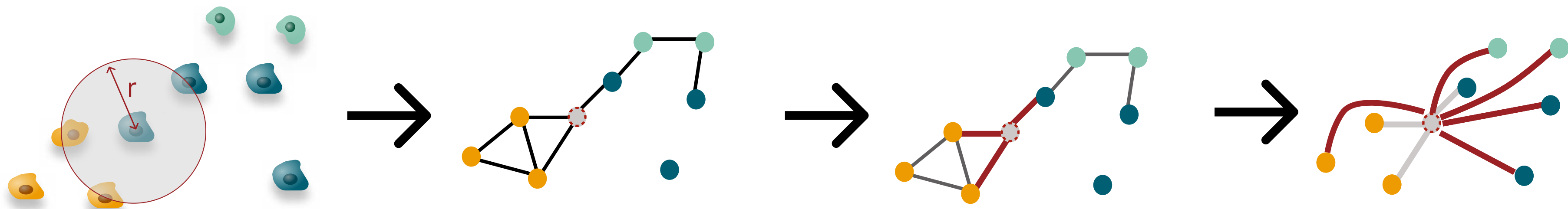
**How to simultaneously
model interactions across
length scales?**



InterScale learns local and global representation



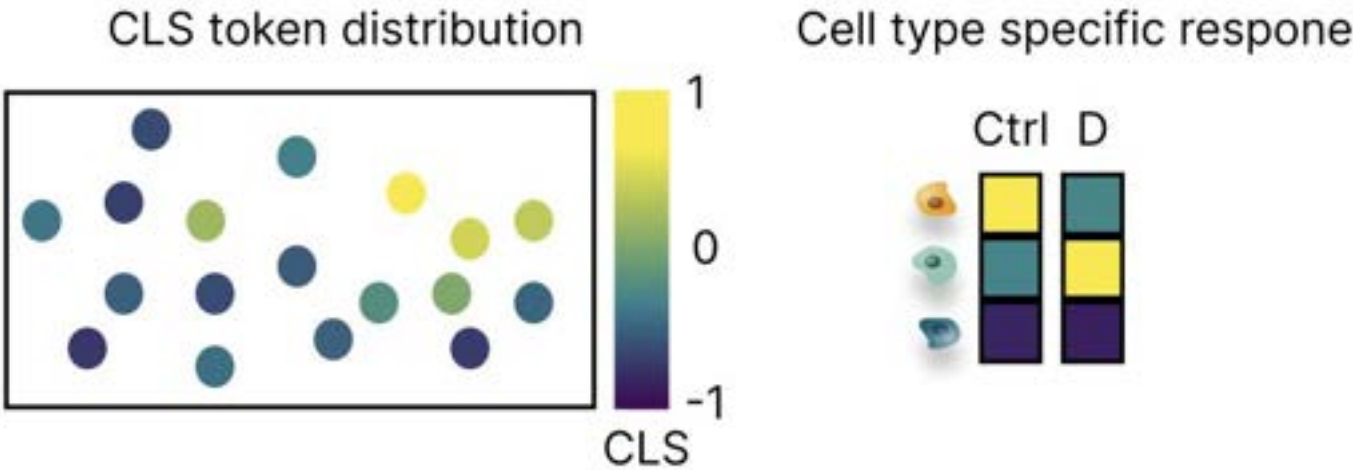
1. Create radius based cell graph
2. Mask n nodes
3. Learn **local** cell representation from **neighbours**
4. Learn **global** cell representation from **tissue**
5. Predict gene expression X



Multi-scale analysis

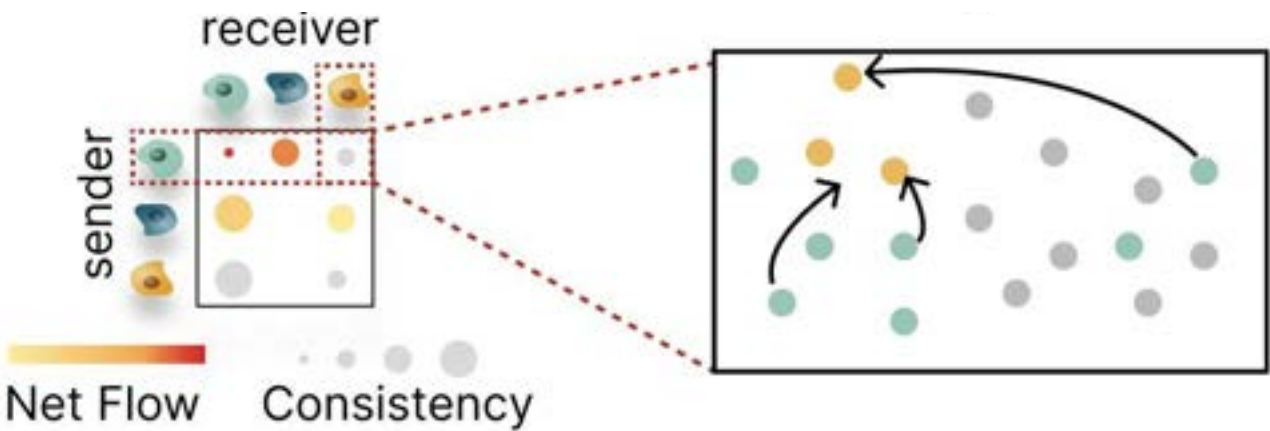
Tissue level

Which cells contain more information for condition prediction?



Cell level

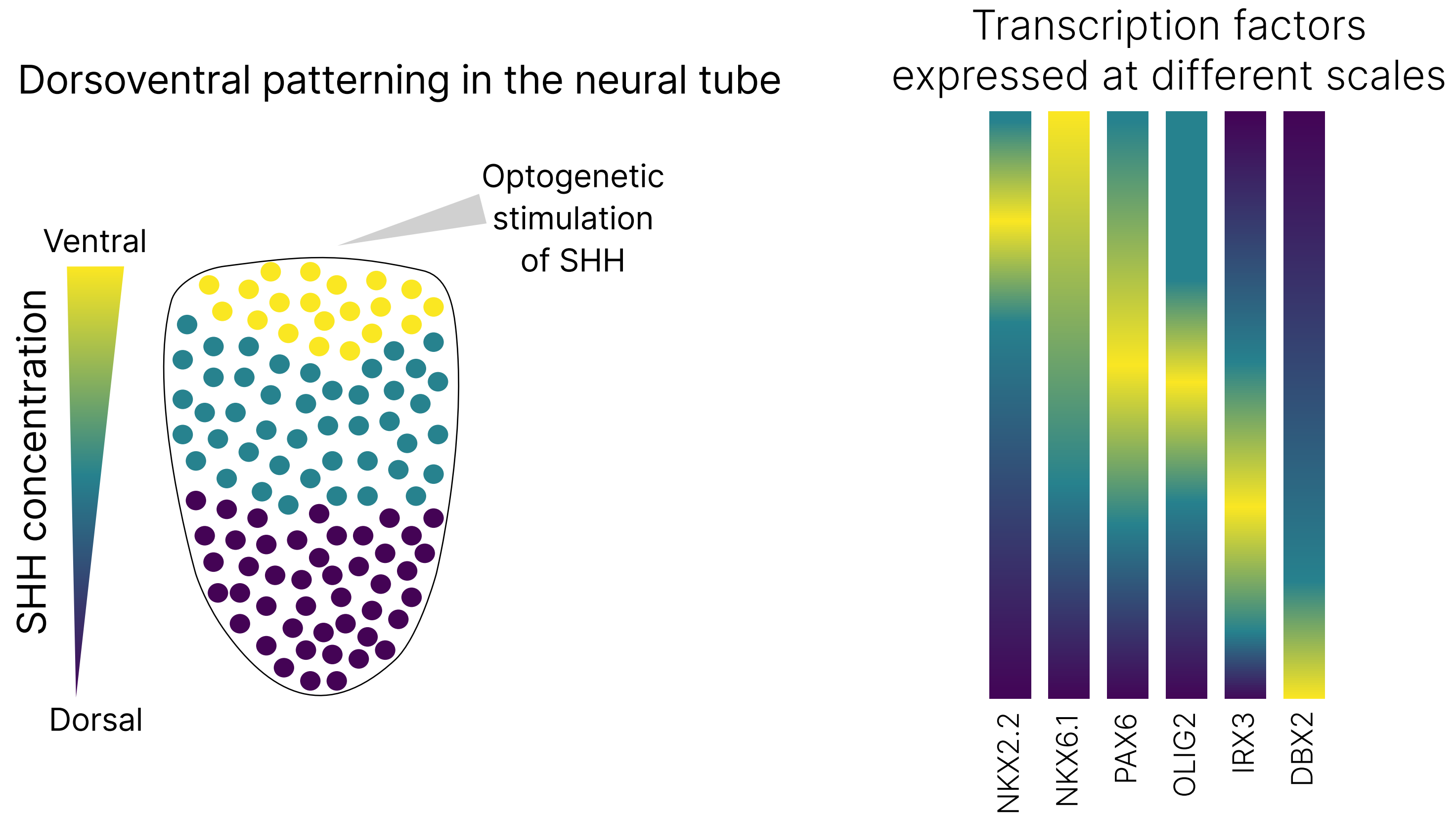
Given a sender and receiver cell (type), what is their interaction strength?



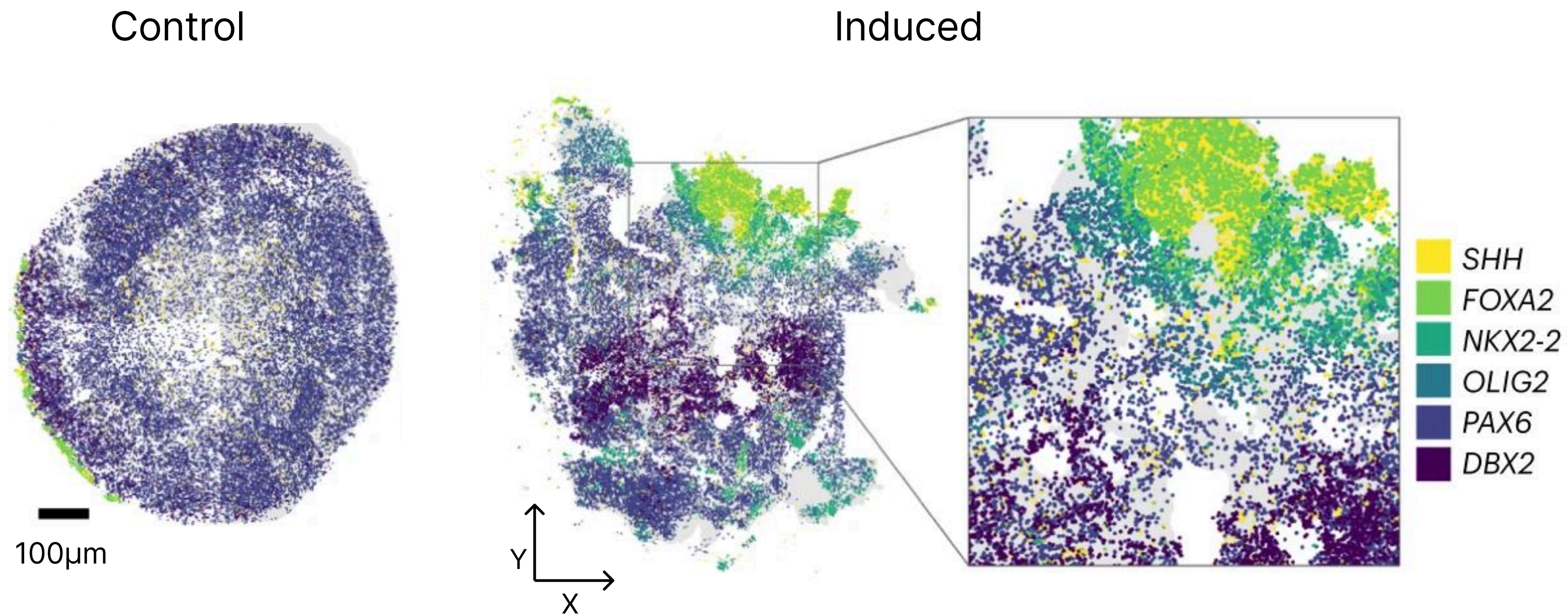
Gene level

Which genes are more relevant local vs global interactions?

Sonic Hedgehog (SHH) creates dorso-ventral patterning

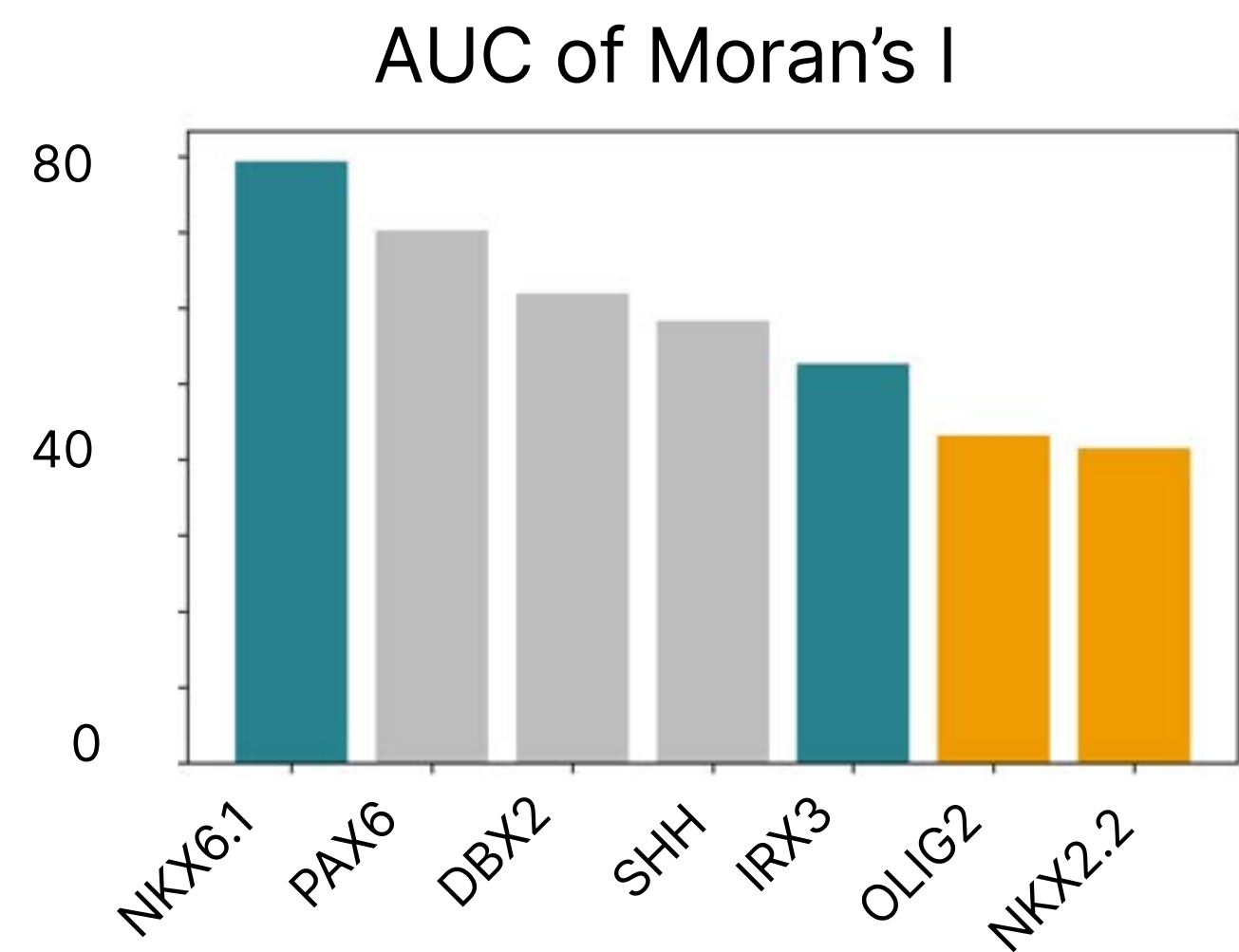
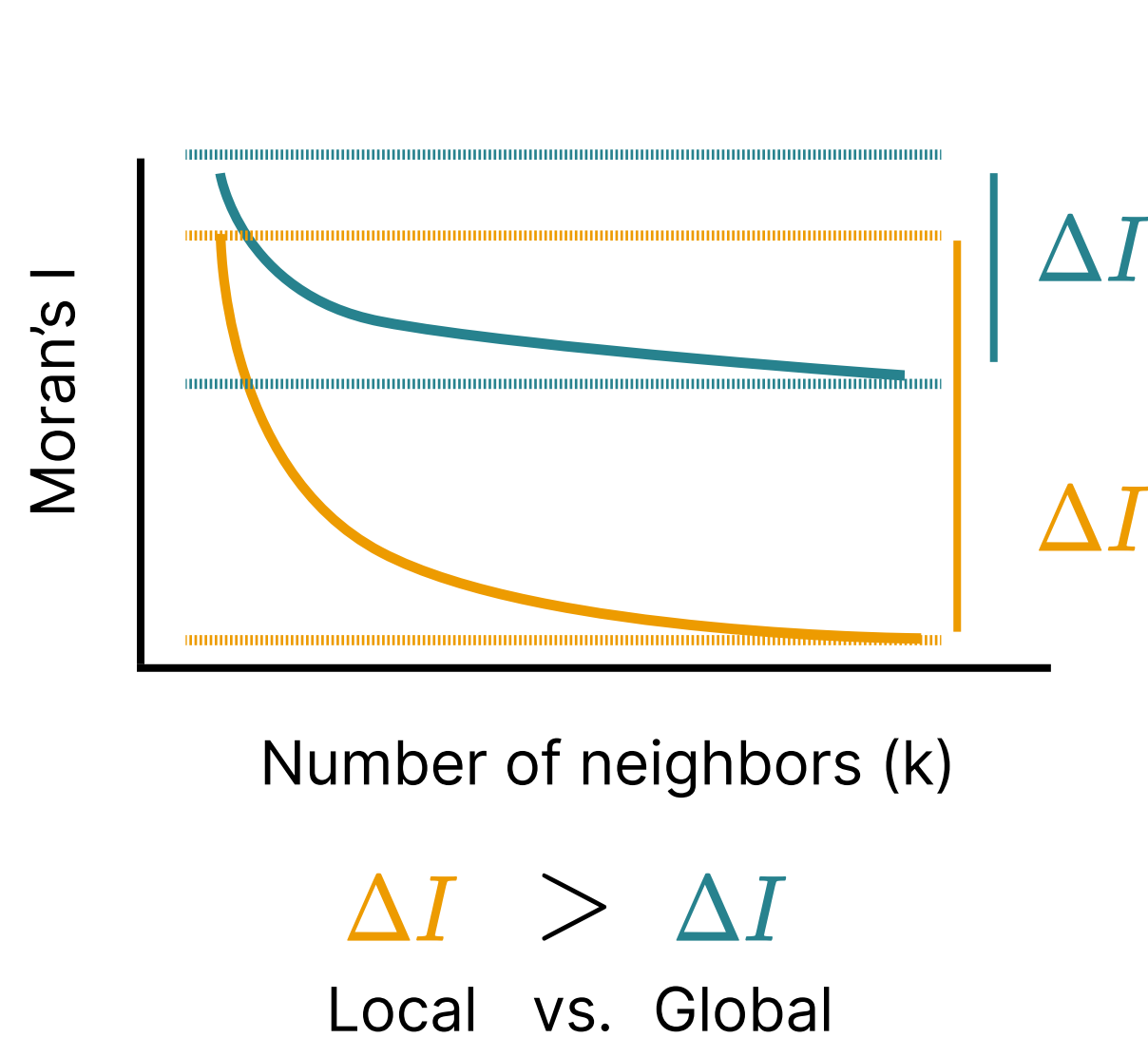
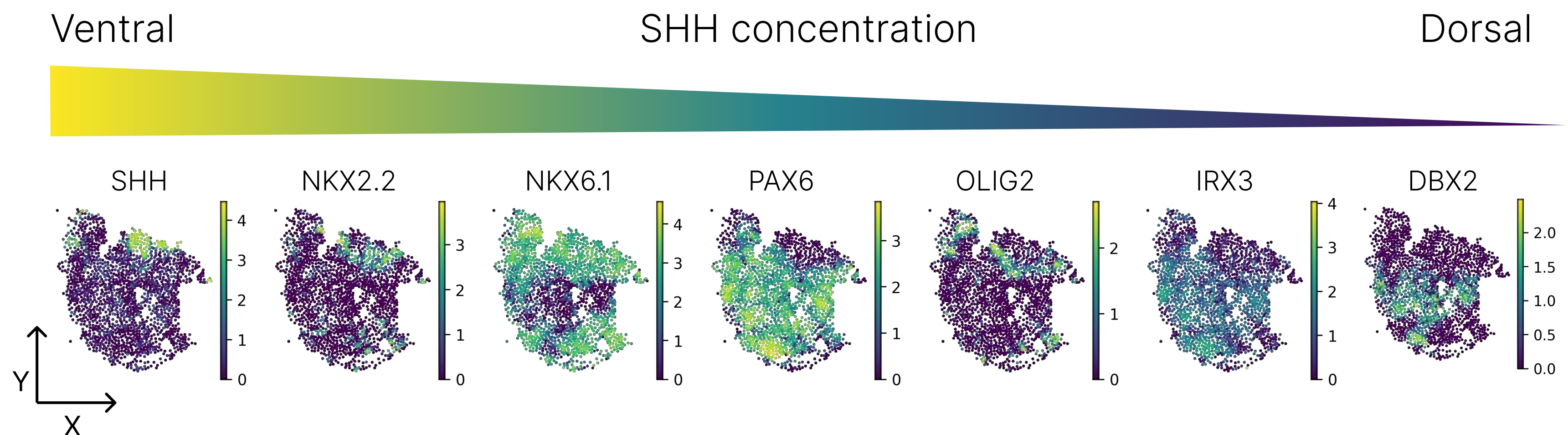


Sonic Hedgehog (SHH) signalling across tissue

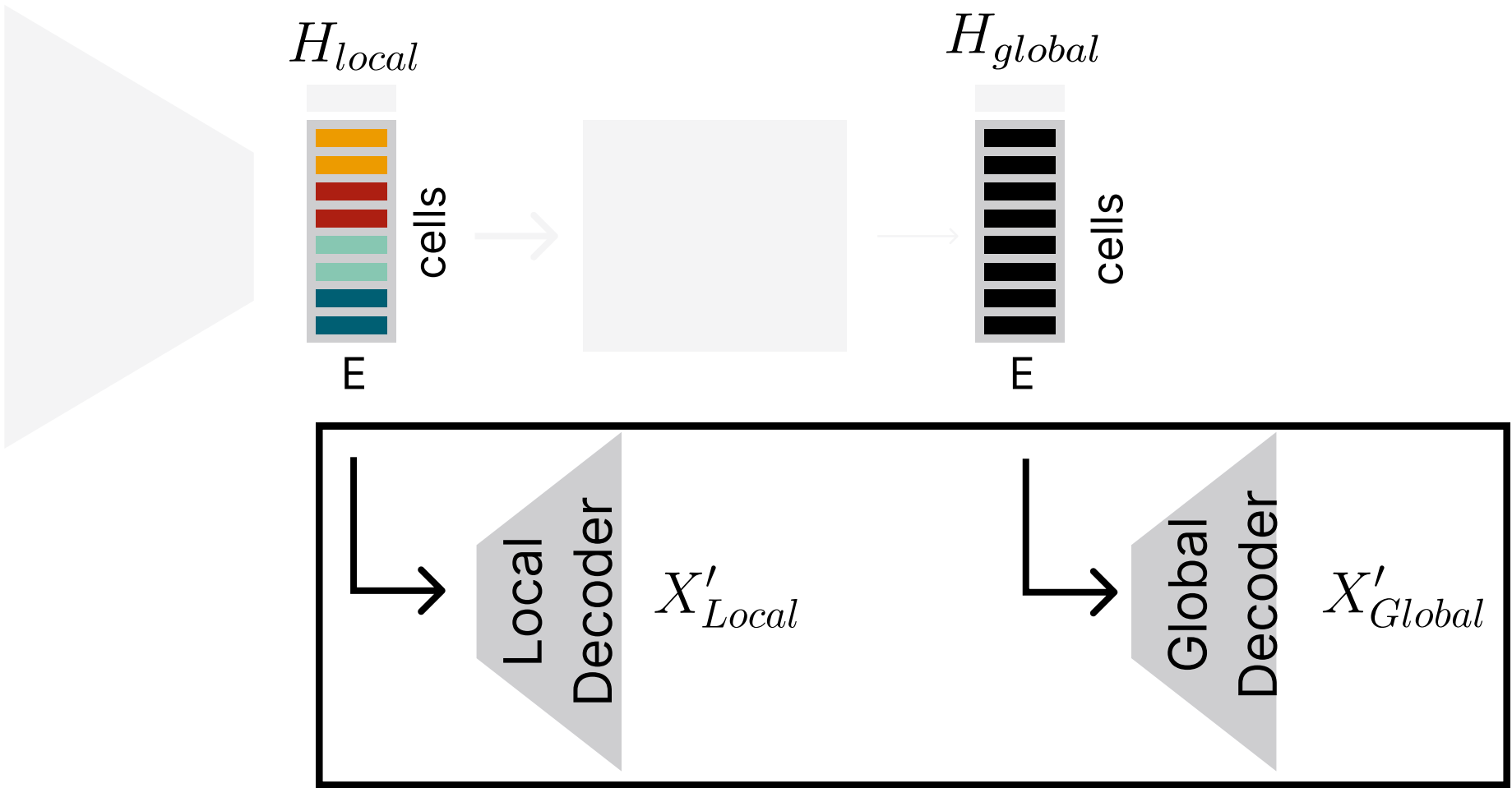


Molecular Cartography

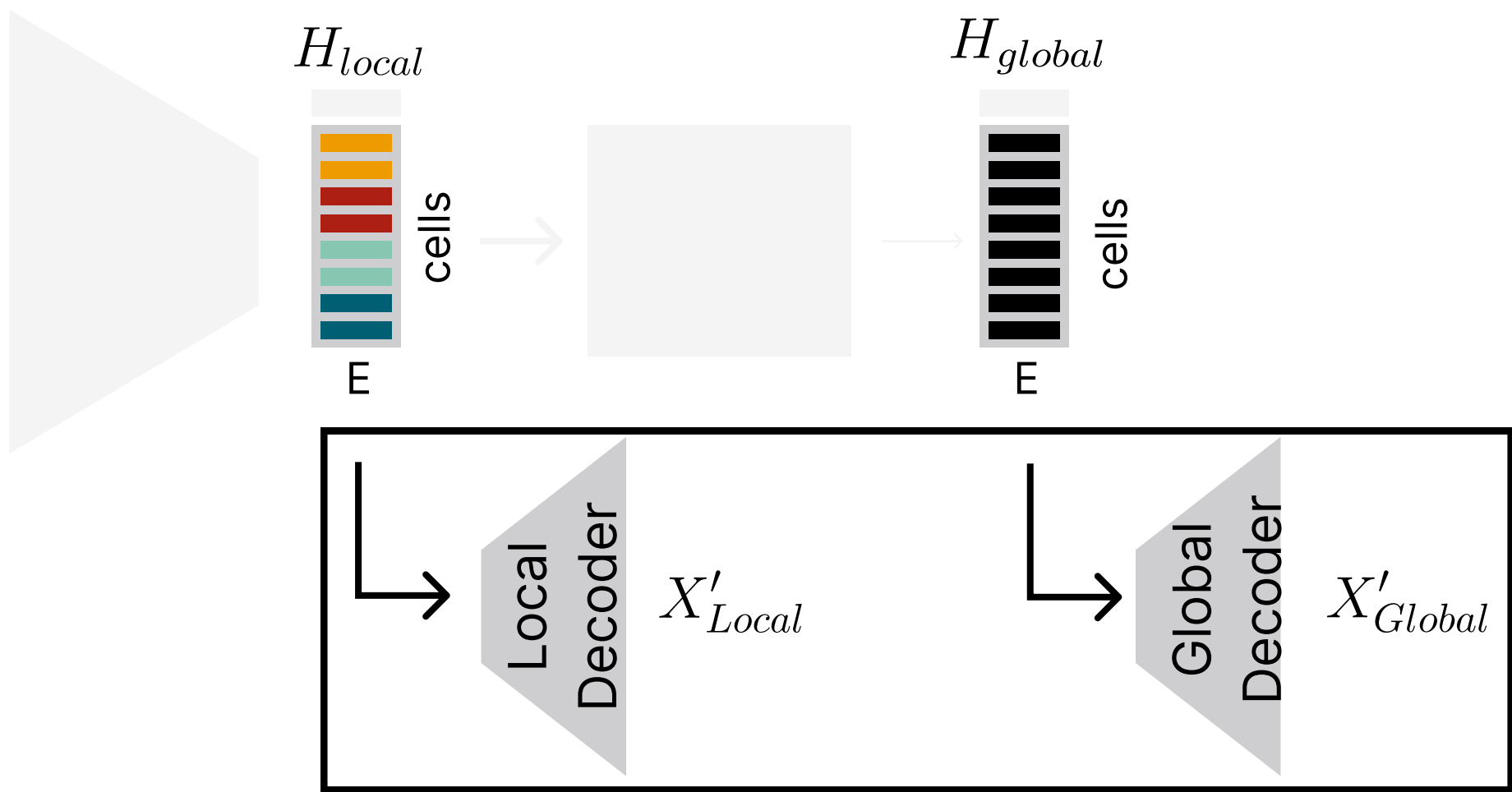
Characterising local vs global gene expression patterns



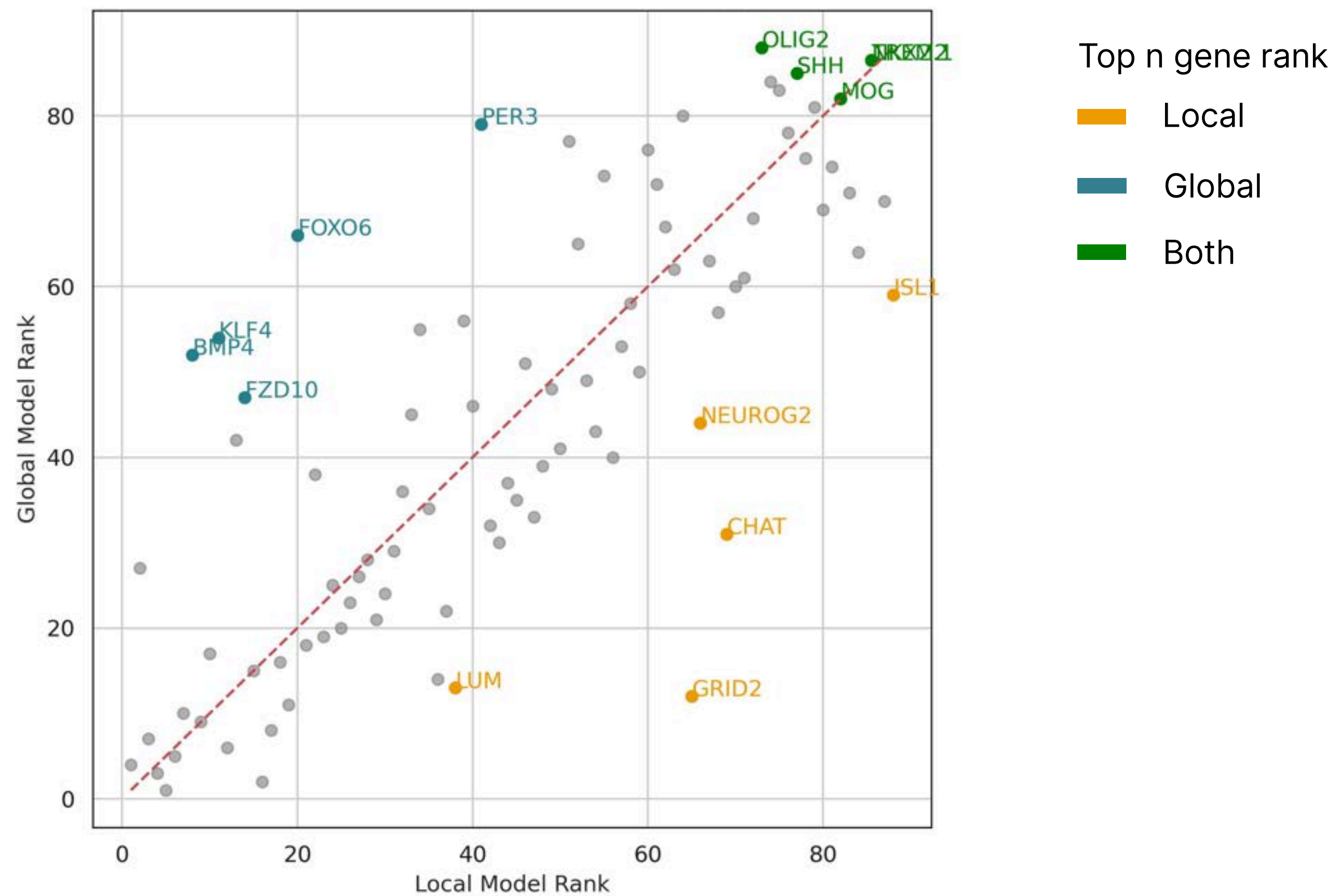
Gene level analysis



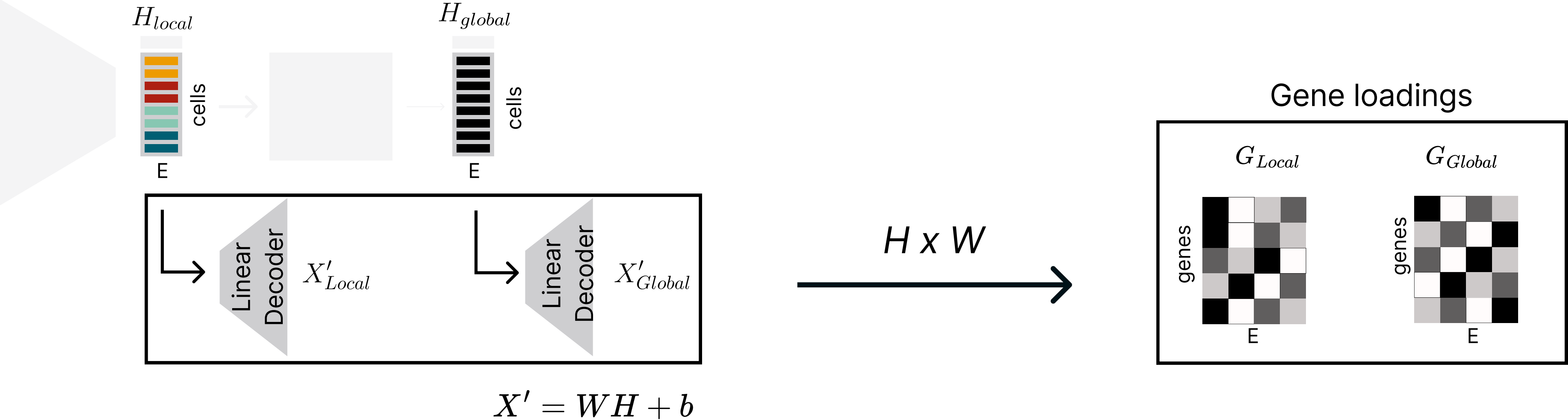
Local and global encodings capture different signals



Gene rank analysis: Local vs Global



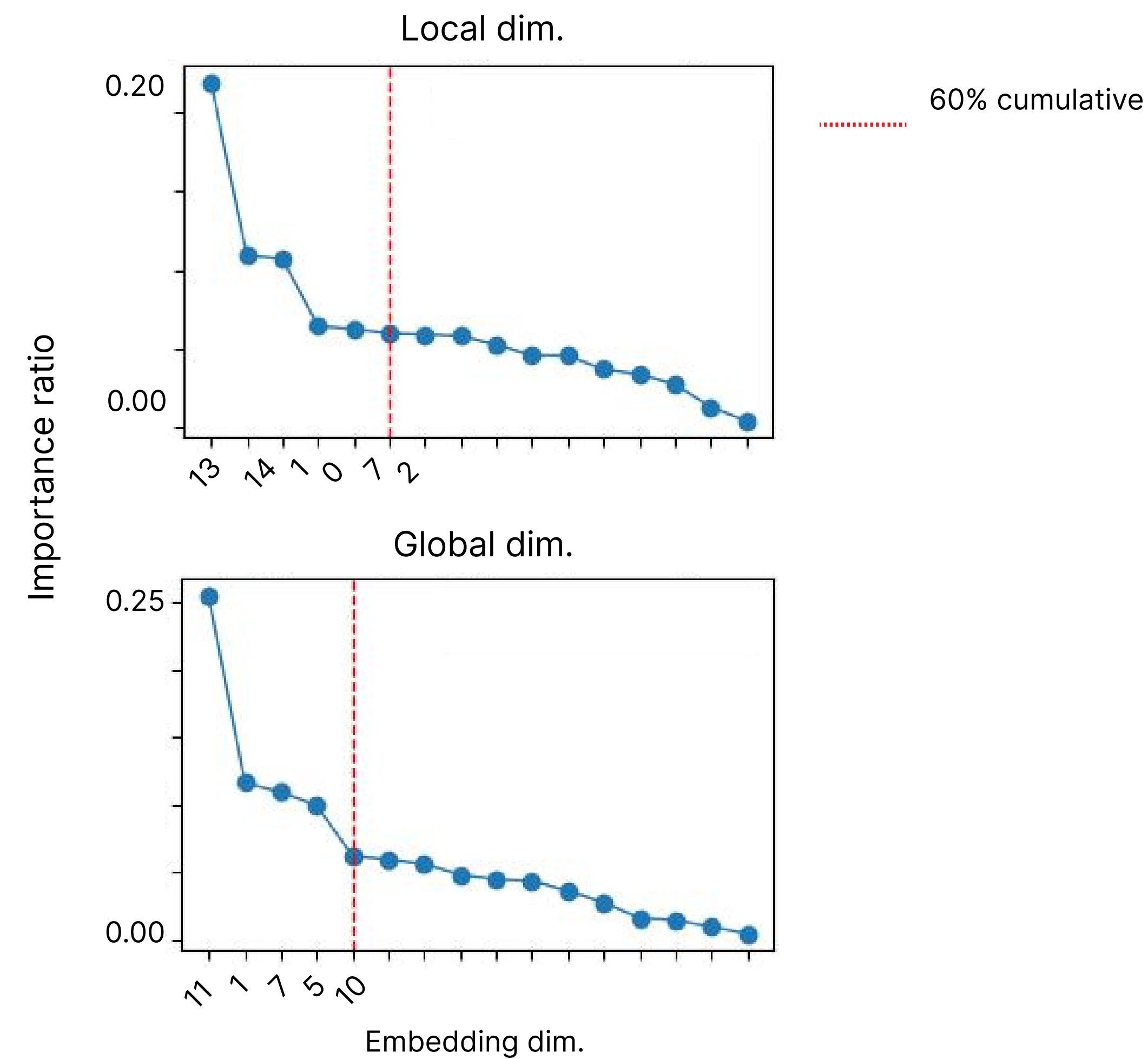
Calculate gene loadings



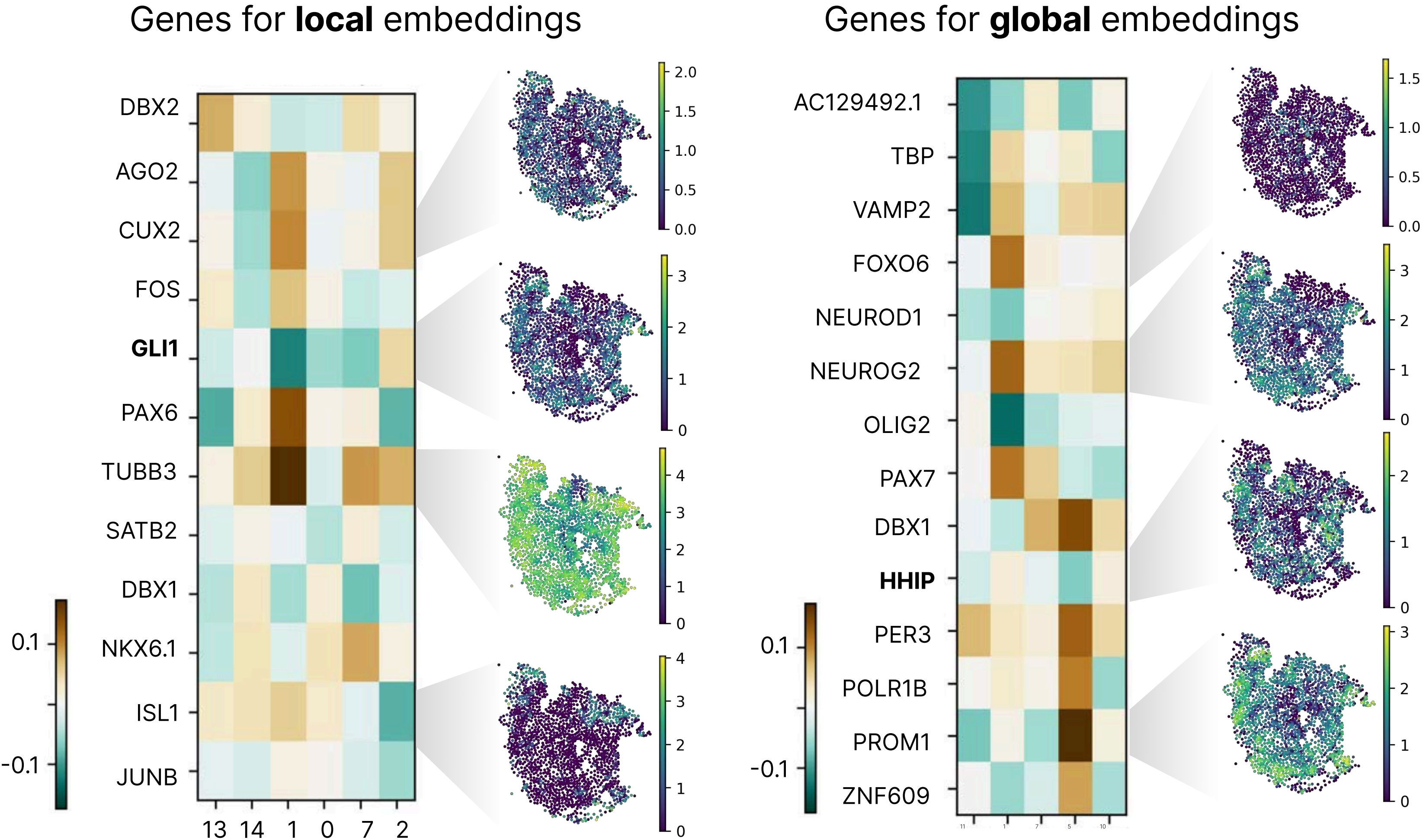
Linear decoders

are directly interpretable.

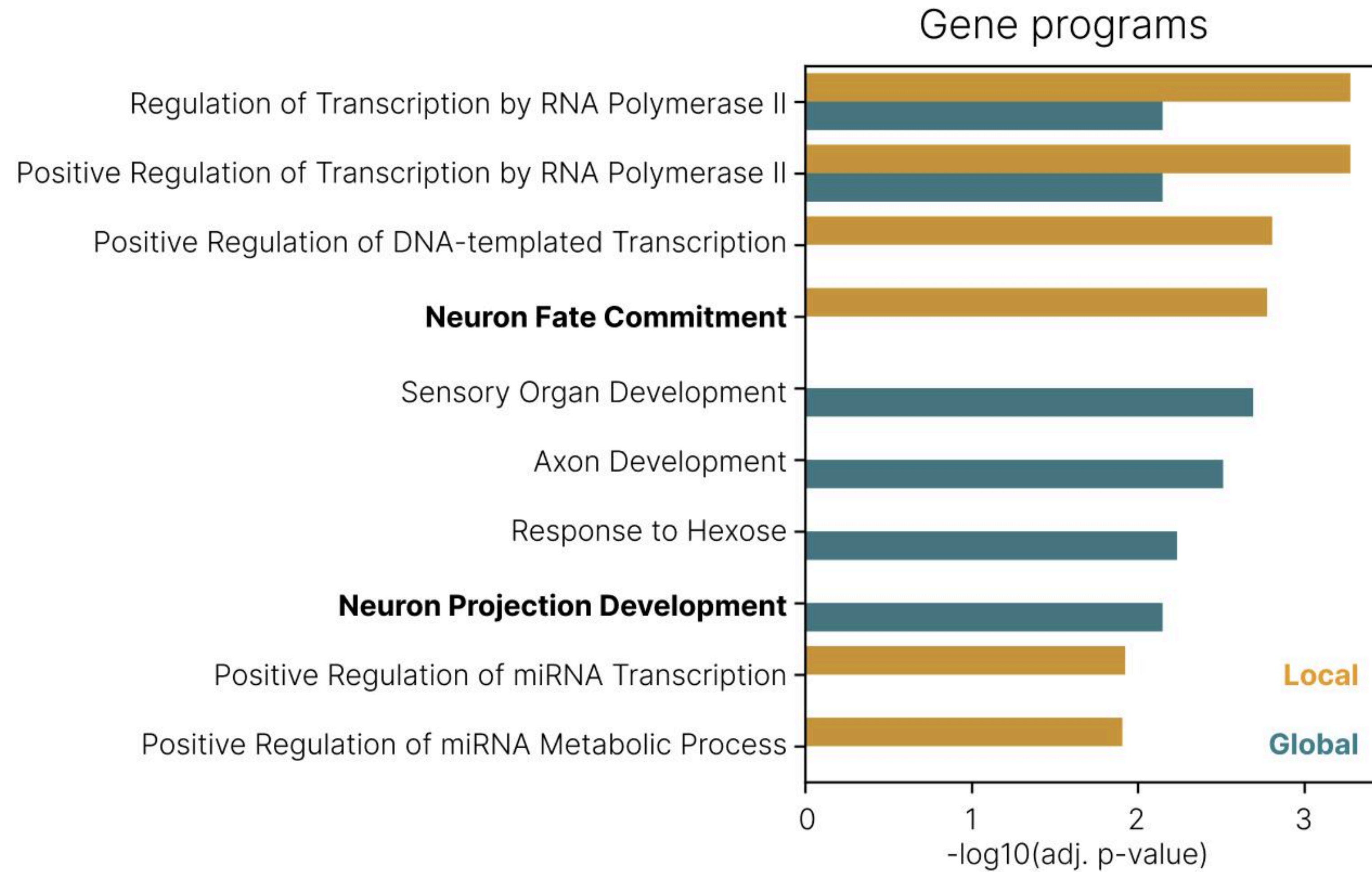
1. Filter embedding dimensions



2. Identify top genes per embedding



3. Gene programs

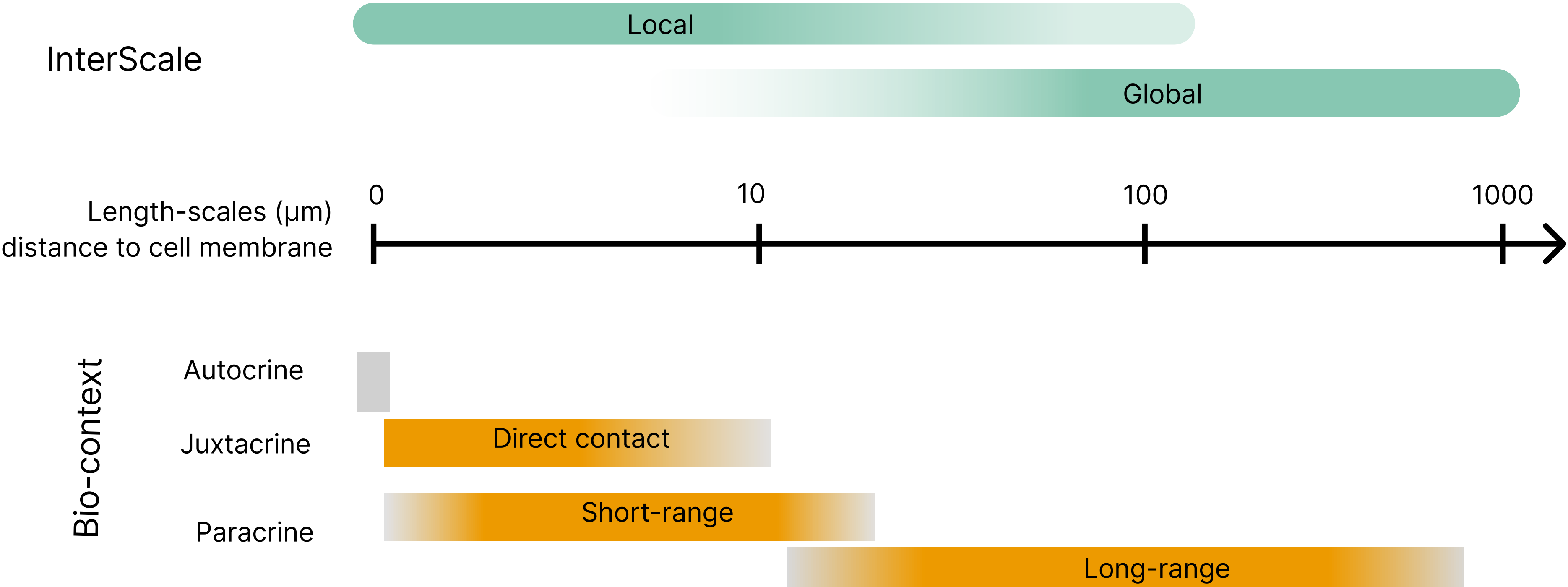


New and ongoing avenues for CCC modelling

Discussion

- different spatial omics technologies require different assumptions

InterScale learns across different length scales

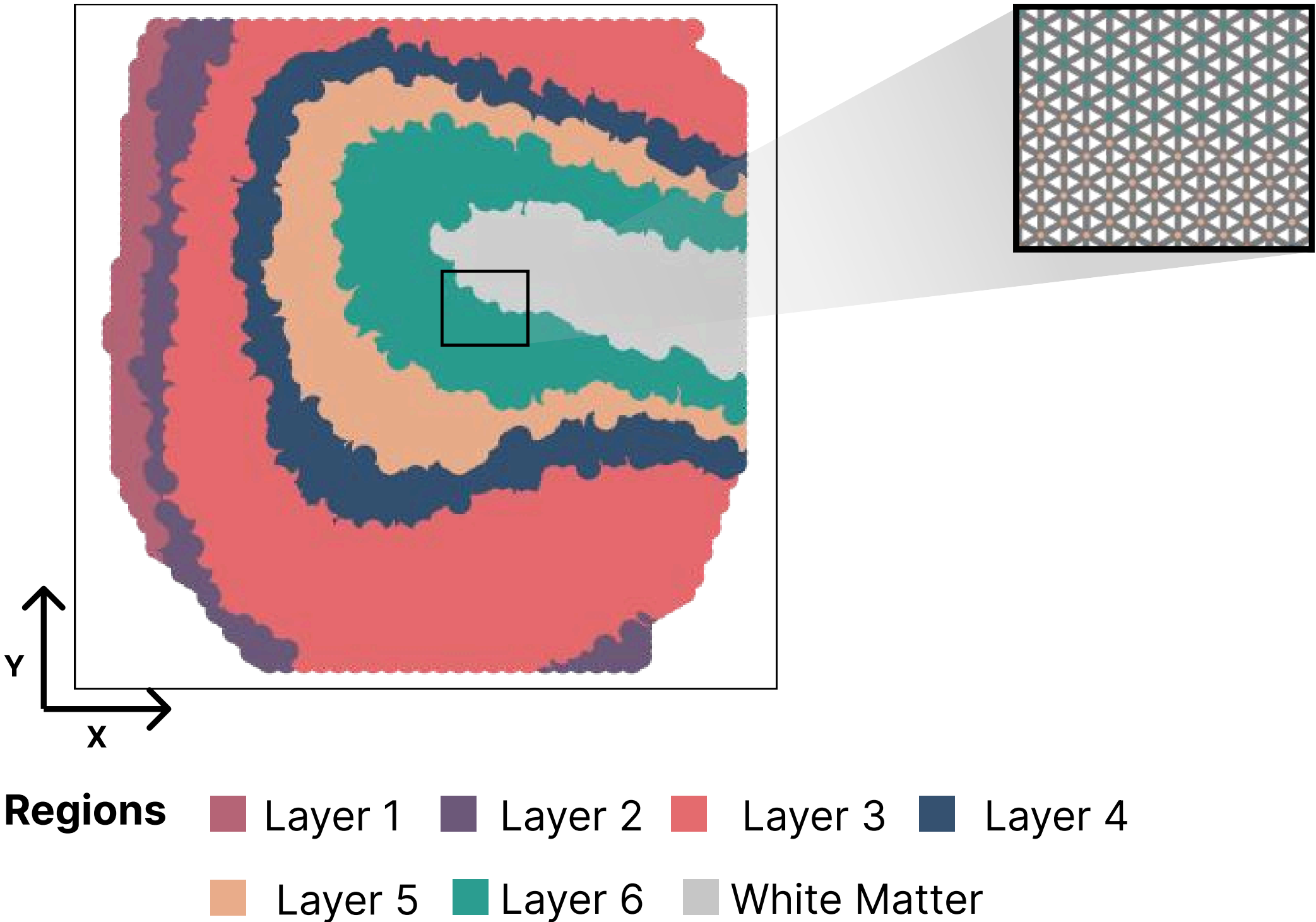


Different ST technologies learn different interactions

10X Visium (sequencing-based)

Alzheimer disease (brain - human)

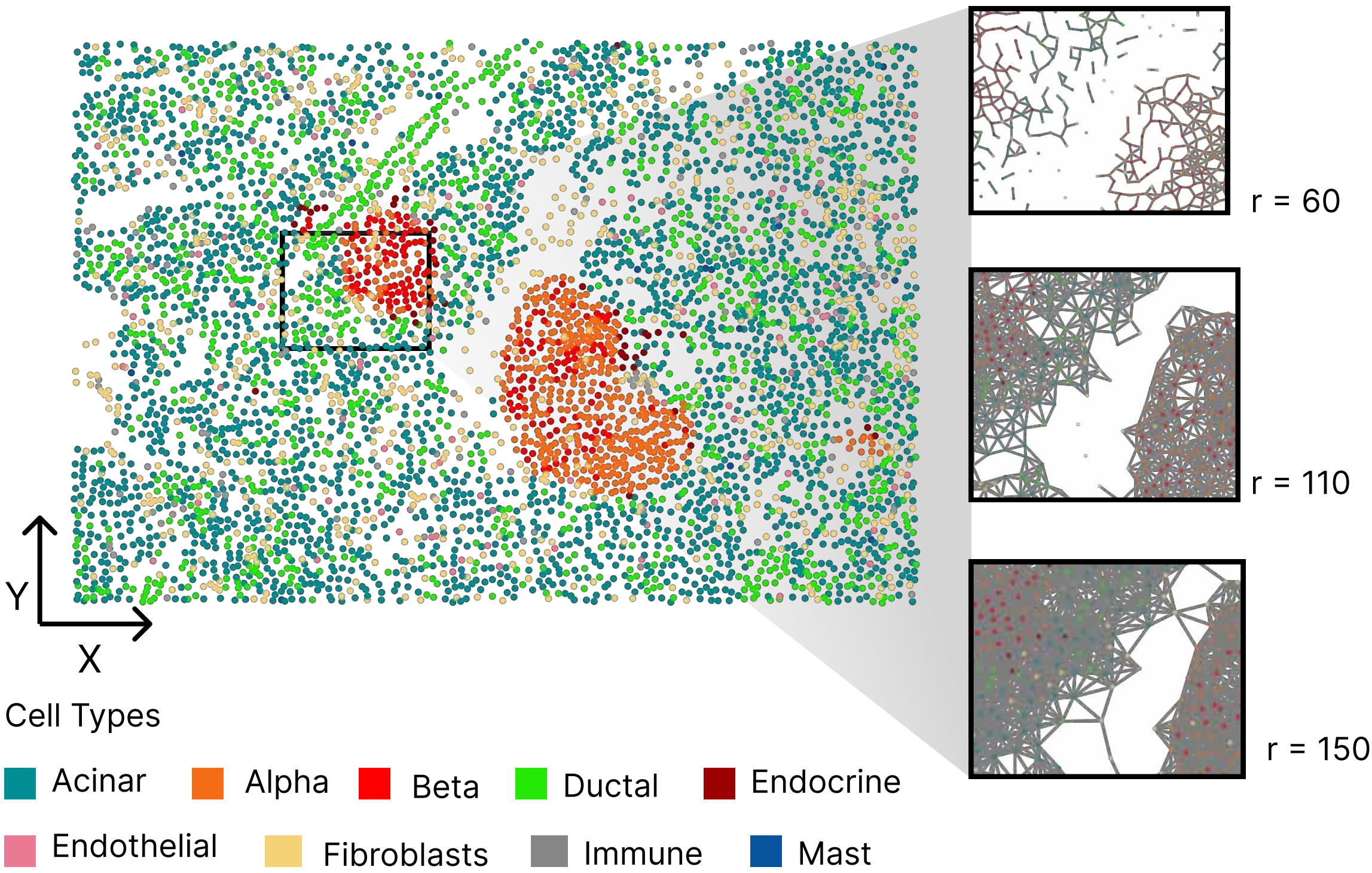
Neighbor connectivity



CosMx (imaging-based)

Type 1 Diabetes (pancreas - human)

Radius connectivity

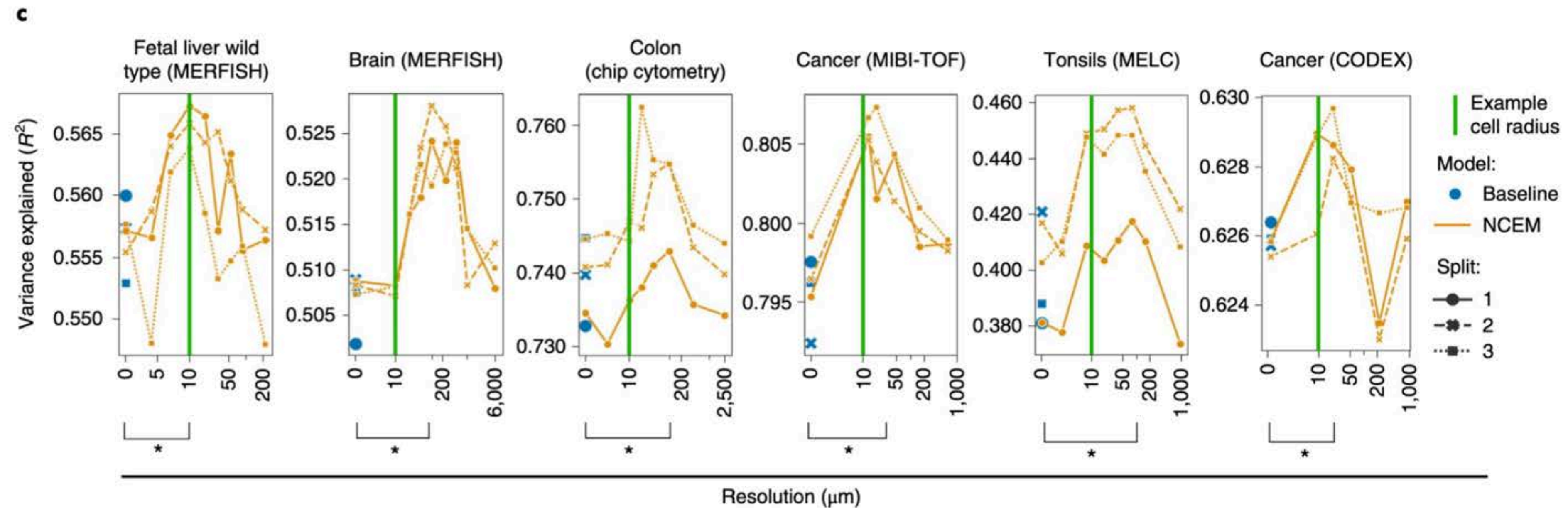


Chen, W.-T. et al. Spatial Transcriptomics and In Situ Sequencing to Study Alzheimer's Disease. Cell 182, 976-991.e19 (2020).

Melton, R. et al. Single-cell multiome and spatial profiling reveals pancreas cell type-specific gene regulatory programs of type 1 diabetes progression. Science Advances 11, eady0080 (2025).

Discussion

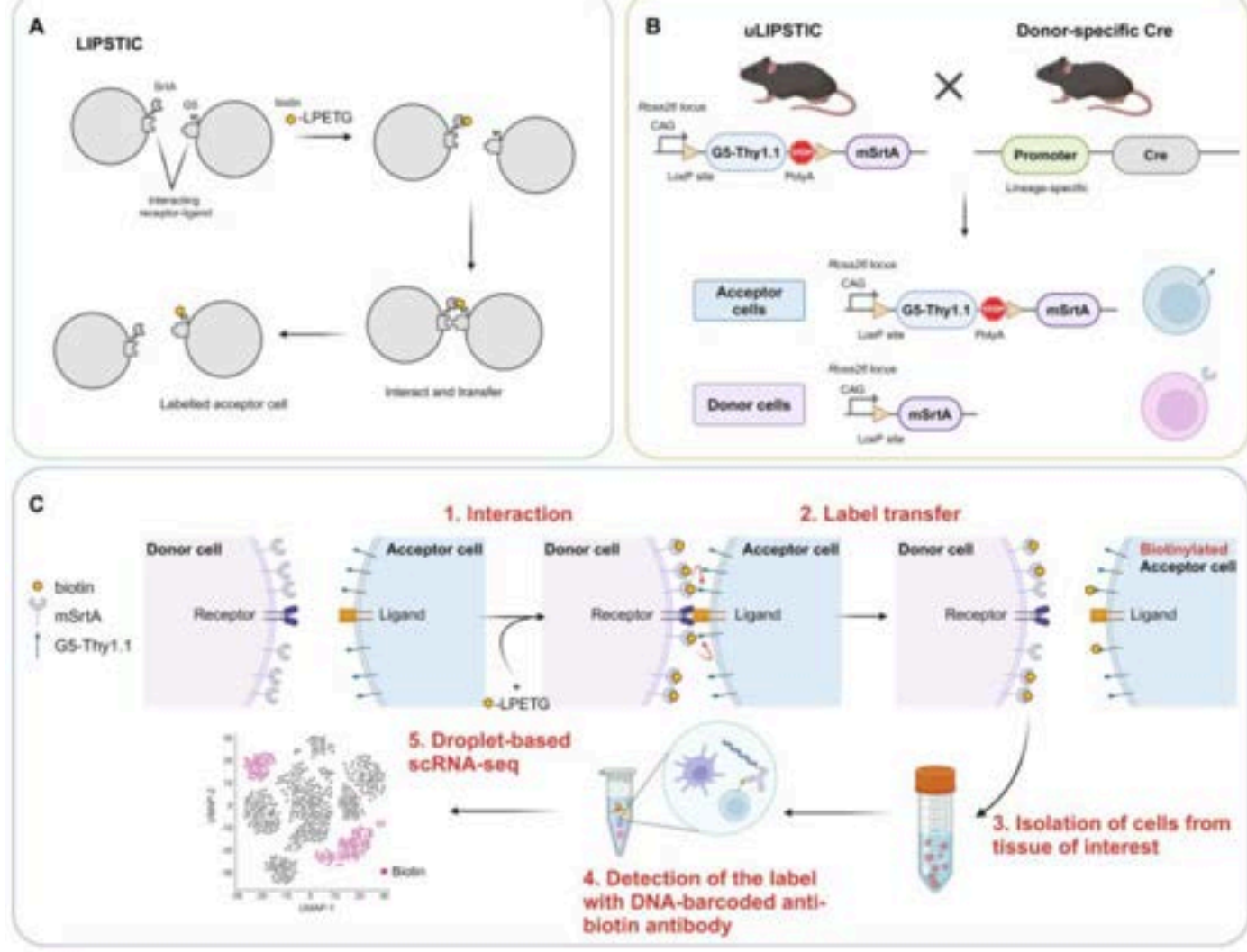
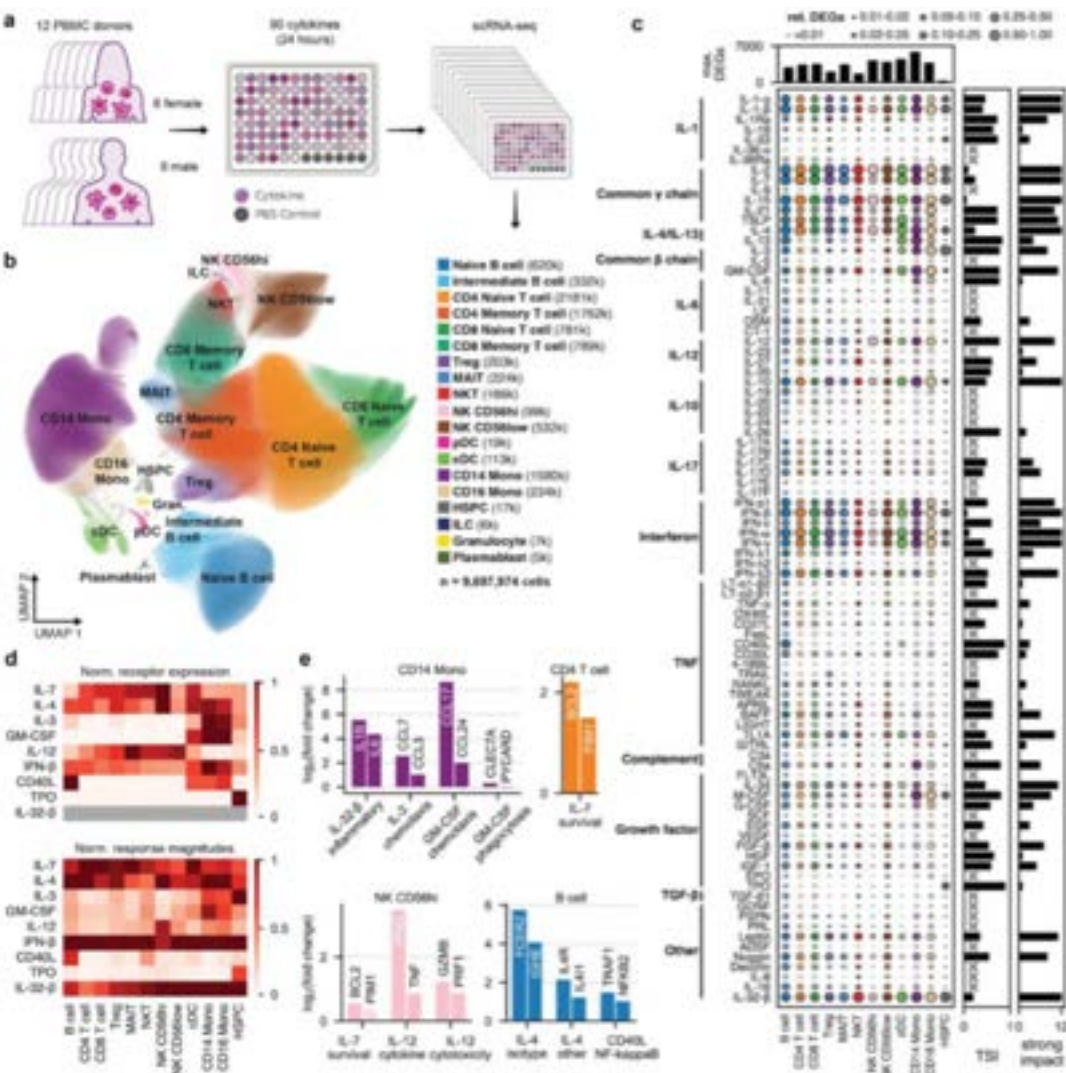
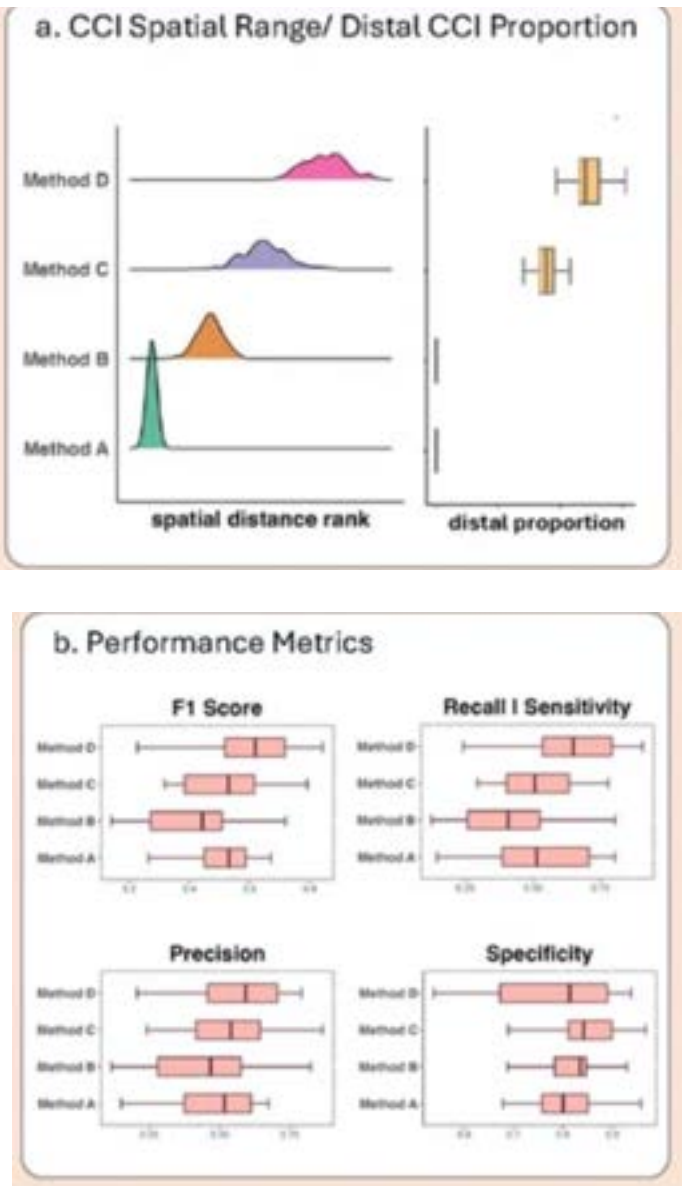
- different spatial omics technologies require different assumptions
- co-localization does not equal communication
- most methods rely on some type of kNN or radius definition which is not trivial



Fischer, D. S., Schaar, A. C. & Theis, F. J. Modeling intercellular communication in tissues using spatial graphs of cells. *Nature Biotechnology* 41, 332–336 (2023).

Discussion

- different spatial omics technologies require different assumptions
- co-localization does not equal communication
- most methods rely on some type of kNN or radius definition which is not trivial
- on-going benchmarking and ground truth dataset creation



- Cui, A., Huang, T., Li, S. et al. Dictionary of immune responses to cytokines at single-cell resolution. *Nature* **625**, 377–384 (2024). <https://doi.org/10.1038/s41586-023-06816-9>
- Oesinghaus, L. et al. A single-cell cytokine dictionary of human peripheral blood. *Res Sq* rs.3.rs-8337240 (2025) doi:10.21203/rs.3.rs-8337240/v1.

Nakandakari-Higa S, Walker S, Canesso MCC, et al. Universal recording of immune cell interactions in vivo. *Nature*. 2024;627:399-406. <https://doi.org/10.1038/s41586-024-07134-4>

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scverse GSCN course



Sara Jimenez

Now: @NVIDIA
Prev.: Postdoc @Theislab



Fabian J. Theis

PI



everyone in the Theislab

**HELMHOLTZ
MUNICH**



Thank you!



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🐙 [FrancescaDr](#)